

Software-as-a-Graph: Heterogeneous Graph Learning for Pre-Deployment Dependability Analysis of Complex Distributed Systems

Abstract

Assessing distributed system dependability before deployment is hindered by absent runtime telemetry and static code analysis’s blindness to communication topology. We present Software-as-a-Graph (SaG), a static system analysis framework that constructs typed multigraphs from Architecture-as-Code manifests across five entity types. SaG pairs two parameter-independent pathways: a relation-specific Heterogeneous Graph Transformer (HGT) with Quality-of-Service (QoS) edge encodings to forecast cascading failure blast radii, and an interpretable ISO/IEC 25010 layer attributing fragility to Availability (single points of failure) or Fault Tolerance (cascade hubs) to guide targeted repairs.

Across eleven synthetic scenarios and five open-source reference systems: (1) relation typing gives a small but exceptionally consistent gain over homogeneous learning at matched training sets and model selection — $\Delta\rho = +0.082$ under distribution shift, winning *all twelve* folds ($p = 0.0005$), replicated by the unweighted pair (+0.064, 12/12); (2) against a training-free QoS-weighted centrality baseline the learned model establishes no advantage (+0.058, $p = 0.204$ at fixed configuration; +0.115, $p = 0.052$ under nested selection); (3) on those systems it transfers poorly — $\rho = 0.125$ against 0.516 and 0.511 for two training-free references on identical labels, negative on two of five — yet its own prediction spread flags every one of them, so the failure is detectable at inference time without labels; (4) an HGT forward pass costs 43.7 ms at 2,000 components, the complete gate 0.02–27.4 s. Training a neural model is therefore justified by relational attention, per-relationship criticality, standards-grounded attribution, and this capacity for self-diagnosis — not by ranking accuracy. Findings (1)–(2) predate a repair to the generator’s QoS sampling and await re-measurement.

Keywords: Graph representation learning, Heterogeneous graph neural networks, Distributed systems dependability, Cascading failures, Static system analysis, Explainable AI

1. Introduction

1.1. Motivation

Modern large-scale distributed software systems increasingly rely on asynchronous, event-driven, and publish-subscribe (pub-sub) architectures. Across diverse domains—from autonomous driving (ROS 2 [1]) and enterprise event streams (Apache Kafka [2]) to cyber-physical backbones (DDS [3]), IoT fleets (MQTT [4]), cloud-native microservices [5, 6], and distributed AI/LLM serving clusters—pub-sub decouples producers and consumers in space, time, and synchronization [7]. Components interact indirectly through intermediate message topics and brokers without maintaining direct static references. Furthermore, modern middleware specifications allow engineers to configure deployment-time Quality-of-Service (QoS) policies—such as reliability guarantees, durability, message priorities, and delivery deadlines—to govern how traffic behaves under peak load and network stress.

While this architectural decoupling confers elastic scalability and operational flexibility, it creates a formidable **visibility barrier** for system performance, reliability, and computational sustainability:

- **Indirect Failure and Degradation Pathways:** In traditional synchronous architectures (e.g., RESTful HTTP or gRPC), component interactions follow explicit caller-callee invocation paths. In asynchronous pub-sub and event meshes, publishers and subscribers share no direct references.

Cascading failures, queue head-of-line blocking, and backpressure propagate across hidden logical paths spanning brokers, shared topics, colocated execution nodes, and shared libraries.

- **Distinct Degradation Mechanisms:** Disturbances in complex distributed systems do not propagate in a uniform manner. They manifest either as *sequential cascades* (e.g., a slow subscriber causing broker queue saturation and upstream backpressure) or as *simultaneous blast radii* (e.g., a shared runtime library crash, memory exhaustion, or host machine outage instantly disabling multiple colocated services). Conventional architectural diagrams and static call graphs fail to represent these multi-layer dependencies.

Addressing these architectural vulnerabilities is most effective and cost-efficient **prior to deployment**, during design and Continuous Integration / Continuous Delivery (CI/CD), adhering to established foundational principles of dependable computing [8]. However, at design and build time, **no runtime telemetry, distributed tracing, or operational logs exist**. Consequently, software architects, performance engineers, and Site Reliability Engineers (SREs) face two fundamental questions without operational data:

1. *Which components, message topics, and communication links are systemically critical to system dependability and performance?*
2. *Why are they critical, and what specific architectural repair (such as replicating a message broker, decoupling an over-subscribed topic, or sandboxing a shared library) will most effectively eliminate that risk?*

The same questions matter for performance and for computational cost: cascading failures that reach production drive restart loops, retry storms and failover thrashing that consume capacity without completing work. Design-time hardening is the cheapest point at which to remove those defects. We flag at the outset that this motivation is broader than what this paper measures — we predict failure impact and report analysis cost, and we neither predict a latency or throughput quantity nor measure energy (Sections 7.6 and 8.2).

1.2. Problem Statement: The Architecture–Code Gap and the Black-Box AI Challenge

We formulate pre-deployment dependability and performance analysis around two distinct, complementary tasks:

1. **Failure-Impact Forecasting (Predictive Pathway) — the primary task.** We forecast dynamic cascading failure blast radii and identify critical components using a data-driven, relation-specific model over learned topological representations. Closed-form topological metrics capture broad connectivity cheaply; whether they also resolve multi-hop, relation-dependent cascade spread across heterogeneous channels is an empirical question, which we test directly against such a baseline (Section 7.2). Our predictive pathway is trained and evaluated against independent simulation ground truth as a ranking and critical-set identification model.
2. **Explainable Criticality Attribution (Explanation Layer) — what a rank alone cannot say.** A ranked shortlist indicates *where* risk lies, but not *how to fix it*. We therefore pair the predictor with an interpretable structural quality profile grounded in ISO/IEC 25010 [9] and ISO/IEC 25019 [10]. This layer diagnoses the *qualitative root cause* of vulnerability—distinguishing, for instance, an unreplicated single point of failure from a high-coupling maintainability bottleneck—to guide concrete repairs. It serves strictly as an attribution model, not a ranking model.

This separation is architectural rather than merely presentational: both pathways operate on the same graph but share no parameters, and neither is trained on the other’s output. The coupling term that could connect them is disabled by default and reported only as an ablation (Section 4.2). Maintaining this independence allows SaG to identify components that are structurally central yet operationally low-impact—a nuanced diagnosis unattainable by either pathway alone.

Existing software engineering approaches fail to bridge what we define as the **Architecture–Code Gap**: *a distributed system can have pristine, bug-free source code within each individual service, yet remain fragile*

to catastrophic global outages caused by hidden architectural single points of failure (SPOFs) or mismatched middleware Quality-of-Service (QoS) contracts. Although classical architecture evaluation methods such as the Architecture Tradeoff Analysis Method (ATAM) [11, 12] identify architectural risks, and software studies analyze architectural technical debt [13] and bad smells [14], they rely on manual stakeholder elicitation rather than quantitative structural learning. Prevailing automated paradigms each leave part of this gap unaddressed. Static code analysis [15, 16, 17, 18] inspects complexity, cohesion and coupling *within* a service but cannot observe message queues or cross-host propagation. Runtime chaos engineering [19] injects real faults but needs a provisioned cluster, carries operational risk, and arrives after the architecture is fixed. Homogeneous centrality metrics [20, 21, 22, 23] flatten the system into an untyped graph in which a message topic, a shared library and an execution host are indistinguishable. Section 2 develops each of these in turn.

Furthermore, while machine learning has demonstrated remarkable success across software engineering, contemporary AI approaches applied to system dependability often function as uninterpretable black boxes. Deep neural models frequently output scalar risk scores or latent embeddings without providing transparent, actionable rationales for their predictions. In mission-critical software engineering, an opaque risk score is inadequate: developers and SREs cannot refactor code or reconfigure infrastructure without understanding *why* a component is vulnerable and *which* architectural mechanism is compromised.

1.3. The Software-as-a-Graph (SaG) Approach

To bridge the Architecture–Code Gap while overcoming the black-box AI challenge, this work introduces **Software-as-a-Graph (SaG)**, an AI-driven pre-deployment **Static System Analysis (SSA)** framework. SaG ingests Architecture-as-Code manifests and executes a four-stage pipeline:

1. **Typed Multigraph Formulation:** SaG models the distributed architecture as a typed, directed multigraph over five core entity types: Applications, Brokers, Topics, Execution Nodes, and Shared Libraries (Section 3.1).
2. **QoS-Aware Logical Dependency Projection:** Using six formal projection rules, SaG derives a semantic `DEPENDS_ON` dependency layer that captures both sequential cascades (via topics and brokers) and simultaneous blast radii (via shared libraries and node colocation), weighted by declared QoS contracts (Section 3.2).
3. **Heterogeneous Graph Learning for Failure Forecasting (Predictive Pathway):** SaG trains a **Heterogeneous Graph Transformer (HGT)** whose relation-specific attention lets a `USES` edge into a shared library propagate differently from a `PUBLISHES_TO` edge into a topic. It forecasts cascading blast radii, ranks critical components, and outputs per-relationship criticality alongside multi-task quality outputs (Section 4).
4. **Explainable Quality Attribution (Explanation Layer):** To explain *why* a flagged component is critical, SaG combines code-level SCA metrics with topological properties into a deterministic **Reliability–Maintainability (RM)** attribution model (Section 5). Reliability decomposes into **Fault Tolerance** (error propagation depth) and **Availability** (single-point-of-failure exposure), pointing to distinct repairs. Because it is a linear, propagation-free aggregate by design, it explains *why* a component is vulnerable rather than how far a cascade travels; its standalone rank correlation is correspondingly modest (Section 7.2).

To ensure methodological rigor, SaG enforces a strict **input–label independence guarantee**: learned models and attribution baselines operate exclusively on the analytical graph G_{analysis} , while ground-truth failure impacts are generated by independent discrete-event simulators operating on the raw structural topology $G_{\text{structural}}$ (Section 4.4).

Figure 1 shows how the two pathways relate. The predictive pathway is the primary one and the only pathway validated against the simulation oracle — the oracle scores rankings, which a quality profile is not — and that oracle is strictly an offline training-and-validation component, never a dependency of online inference. The explanation layer then characterises what the predictor flagged and the remediation that implies. The single link between them is triage rather than data flow: the architect applies the explanation to whatever the predictor ranked. The remediation guidance closes a loop of its own, in which each candidate edit is

re-simulated on its own mutated copy of $G_{\text{structural}}$ and kept only if it beats the simulator’s seed-to-seed noise.

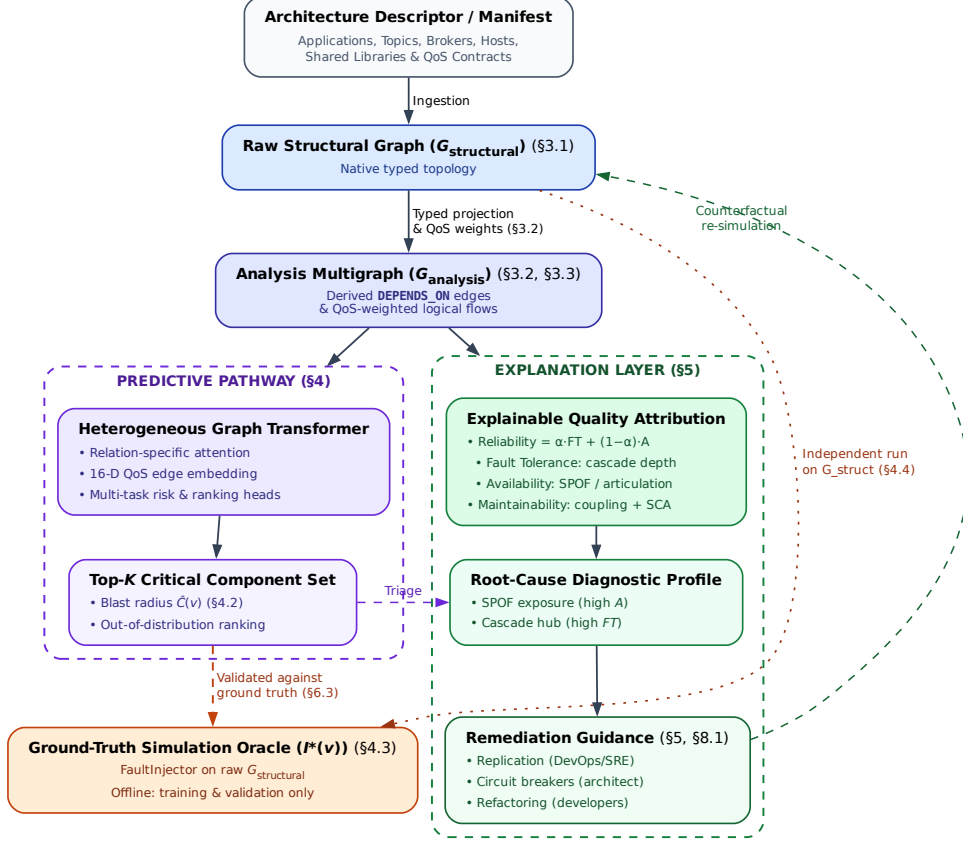


Figure 1: End-to-end architecture of the SaG framework. A shared front end (manifest ingestion $\rightarrow$ typed multigraph $\rightarrow$ QoS-weighted DEPENDS_ON projection $\rightarrow$ typed node features) feeds two pathways that share no parameters: the predictive pathway (Section 4), which emits a ranked critical set and per-relationship criticality, and the explanation layer (Section 5), which emits a standards-grounded quality profile. The simulation oracle scores only the former and runs on $G_{\text{structural}}$ alone.

Rationale for Graph Learning vs. Direct Simulation. Since discrete-event simulation $I^*(v)$ defines ground-truth criticality here, it is fair to ask why train a graph model at all rather than run simulation sweeps or closed-form heuristics. Four practical reasons motivate the design, and Section 7.2 tests the last of them adversarially. Message passing generalizes across labelled and unlabelled entities alike, which mattered more when it was written than it does now: the injector originally reached only active application processes, leaving 30–47% of components per system (topics, hosts) without direct labels, and it has since been extended to express Topic and host-Node failure directly (Section 8.3). We record the change rather than retain the argument, because the coverage gap it appealed to is closed on this corpus; what survives is the weaker claim that a trained model still scores entity types no simulator sweep was run for. Cascade simulation is stochastic and seed-sensitive (label standard deviation reaches 0.416), while a trained model learns a smooth, threshold-marginalized surrogate that re-scores an already-analysed architecture cheaply. Neither a simulator nor an unaugmented GNN returns a root cause in standardized quality terms: knowing that a subscriber lost its feed does not distinguish an unreplicated single point of failure from an error-cascade hub, and

the two demand different repairs. Finally, dynamic simulators need runnable containers or communication harnesses, whereas graph learning scores Architecture-as-Code manifests before any runtime infrastructure exists. Whether these motivations are borne out empirically is a separate question, and Section 7.2 answers it only partly in the framework’s favour.

1.4. Research Questions

This empirical study investigates five research questions:

RQ1 (Predictive Efficacy): *How accurately does heterogeneous graph learning predict cascading failure impact and identify the critical component set, compared with traditional, non-learning network metrics?*

RQ2 (Value of Architectural Typing): *Does modeling distinct entity and dependency types (applications, topics, brokers, hosts, and libraries) yield better failure predictions than homogeneous graph models, and does that advantage hold on architectures the model has never seen?*

RQ3 (QoS Encoding, Calibration and Sensitivity): *How do middleware Quality-of-Service policies, the declared weighting constants of the explanation layer, the choice of simulation oracle, and propagation-threshold and normalization settings affect forecasting performance, stability, and explainability?*

RQ4 (Real-World Generalization): *How effectively does the framework transfer zero-shot to authentic, real-world distributed systems across autonomous driving (ROS 2), cloud-native microservices, smart home IoT, and industrial edge computing?*

RQ5 (Analysis Cost and Computational Sustainability): *What does pre-deployment analysis cost at CI/CD time, which pipeline stage dominates that computational footprint, and does the resulting budget enable sustainable, per-commit architectural gating?*

1.5. Key Contributions

This paper presents four principal contributions:

1. **Heterogeneous Graph Learning for Pre-Deployment Dependability:** A relation-specific Heterogeneous Graph Transformer that forecasts cascading blast radii from Architecture-as-Code alone, with a 16-dimensional edge feature vector carrying 7 QoS dimensions and multi-task heads for component and relationship criticality (Section 4). Our central empirical claim concerns cross-architecture transfer: with matched training sets, depth and model selection, typed learning leads untyped learning by $\Delta\rho = +0.082$ (0.659 vs. 0.577) under inductive distribution shift, winning all 12 folds ($p = 0.0005$; Section 7.3), and the unweighted pair replicates this exactly ($+0.064$, 12/12, $p = 0.0005$). The effect is small — a third of the QoS-weighted baseline’s own $+0.253$ over unweighted centrality — but it is the only comparison in this study that wins every fold, and it holds in-distribution as well ($+0.088$, 6/7, $p = 0.047$), so relational typing is a consistent small benefit in both regimes rather than a purely inductive bias. Because the homogeneous baseline receives the scalar QoS aggregate rather than the full 16-D encoding, this margin could in principle combine typing with per-dimension QoS encoding; an ablation within the typed architecture attributes none of it to the QoS dimensions (-0.005 , $p = 0.68$; Section 7.4.1). Simultaneously, we establish the honest empirical boundary: against an unparameterized QoS-weighted centrality baseline (Topo-QoS), out-of-distribution ranking is not surpassed — $\Delta\rho = +0.058$ ($p = 0.204$) at fixed configuration and $+0.115$ ($p = 0.052$) under nested selection — and the critical-set margin ($F_1@K = 0.439$ vs. 0.375) is not statistically significant either, won in 7 of 12 folds ($p = 0.23$; Section 7.2).
2. **A Formal Typed Architecture Model:** A multigraph representation that derives logical dependencies from physical pub-sub linkages and distinguishes sequential cascade propagation from simultaneous multi-consumer library failures, supplying the typed substrate the predictor consumes (Section 3).

3. **A Standards-Grounded Explanation Layer:** An interpretable Reliability–Maintainability model grounded in ISO/IEC 25010/25019 that turns the predictor’s ranked output into an actionable diagnosis, separating single-point-of-failure exposure from error-propagation depth—two distinct failure modes requiring different repairs (Section 5).
4. **Empirical Benchmark, Real-World Evaluation, and Cost Characterization:** An evaluation across eleven synthetic topologies (2,387 components) and five open-source reference systems (351 components) under strict graph-view separation, establishing both where typed graph learning helps and the boundary where it does not, with a per-stage cost profile locating the pipeline’s cost in its deterministic graph-analysis stage rather than in the learned model (Sections 6–7).

Relationship to the authors’ prior work. An earlier, shorter version was presented at a peer-reviewed conference [24], covering the preliminary typed multigraph formulation and the deterministic quality-attribution model on the synthetic corpus alone. This manuscript is a substantially extended version meeting the JSS extension policy. New here are: the entire predictive pathway (the HGT, its 16-D QoS edge encoding, the multi-task masked-loss heads, and every learned result in Section 7); the inductive LOSO protocol and cross-architecture transfer analysis (Section 7.3); zero-shot evaluation on five open-source reference systems (Section 7.5); the cost characterization answering RQ5 (Section 7.6); the four-oracle convergent-validity taxonomy and graph-view separation (Sections 4.3–4.4); global sensitivity analysis over all ten weight constants (Section 7.4); and the homogeneous-versus-heterogeneous comparison under substrate parity (Section 7.3). Material retained from the conference version is limited to preliminary formalisms in Sections 3 and 5, both restructured and expanded. No companion manuscript from this work is under consideration elsewhere.

1.6. Paper Organization

The remainder of this paper is organized as follows: Section 2 reviews related work on distributed systems dependability, performance engineering, static system analysis, and graph representation learning. Section 3 formalizes the Software-as-a-Graph architectural model, the dependency projection rules, and the typed node features consumed by both pathways. Section 4 presents the Heterogeneous Graph Transformer, its multi-task heads, the simulation oracles that supply its labels, and the input–label independence guarantee. Section 5 introduces the interpretable RM explanation layer. Section 6 describes the experimental setup, benchmark corpus, and evaluation protocols. Section 7 presents empirical results for RQ1–RQ5. Section 8 discusses architectural implications, performance and computational sustainability, threats to validity, limitations, and concluding remarks.

2. Related Work

This work builds upon and connects four foundational research areas: (1) dependability, performance, and sustainability in distributed software systems; (2) static code and system analysis; (3) software quality measurement and multi-criteria evaluation; and (4) graph representation learning and explainable AI (XAI).

2.1. Dependability, Performance, and Sustainability in Distributed Software Systems

The publish–subscribe (pub-sub) and asynchronous event-driven paradigms decouple communicating entities in space, time, and synchronization, enabling elastic scalability and high throughput [7]. Modern middleware standards—such as ROS 2 [1], Apache Kafka [2], DDS [3], and MQTT [4]—govern these exchanges through fine-grained Quality-of-Service (QoS) policies that regulate message durability, transport reliability, priorities, and delivery deadlines. In cloud-native microservice meshes and distributed AI/LLM serving backbones, asynchronous message passing and queueing topologies form the primary communication substrate, directly shaping tail latencies, throughput bottlenecks, and hardware resource utilization.

Prior dependability and performance research has focused predominantly on **runtime mechanisms**, including dynamic consensus protocols, broker clustering, adaptive backpressure throttling, autoscaling, and automated failover. In parallel, **chaos engineering and runtime verification** [19] inject faults or latency into staging or production clusters to observe degradation and recovery. This delivers operational validation

that no static method can match, but it requires a fully provisioned cluster, carries the risk of real service disruption, and consumes substantial cluster time per sweep — so it cannot be applied during architectural design or early CI/CD.

Our work addresses the complementary **pre-deployment phase**: predicting systemic cascading vulnerabilities directly from Architecture-as-Code descriptors before runtime infrastructure is provisioned, enabling architectural gating within CI/CD pipelines without a running cluster.

Architecture-Based Reliability Prediction. Predicting dependability from an architectural description before deployment is not a new ambition, and SaG should be read against the tradition that pursued it analytically. Cheung’s absorbing-Markov-chain model [25] derives system reliability from component reliabilities and a transfer-of-control graph; Goseva-Popstojanova and Trivedi [26] systematize the state-based, path-based and additive families that followed, and Immonen and Niemelä [27] survey the resulting methods from the architectural perspective. Model-driven descendants such as the Palladio Component Model [28] and layered queueing networks [29] predict performance and reliability from parameterized component models with well-understood solution techniques.

These methods are complementary to ours rather than superseded by it, and the distinction is one of required inputs rather than of accuracy. They need per-component failure probabilities, transition probabilities or service demands — parameters that are themselves estimated from operational profiles, measurement or expert judgement, and that are unavailable in the setting we target (a manifest at commit time, with no telemetry). SaG asks a narrower question in exchange: not what the system’s reliability *is*, but which components’ failures would propagate furthest through the declared topology. Where those parameters can be obtained, an analytical model answers a stronger question than a ranking does, and we make no claim to displace it.

Data-Driven Failure Prediction and Root-Cause Analysis in Microservices. A large recent literature localizes faults in microservice systems from operational data. Seer [30] and Sage [31] predict and debug QoS violations from traces and hardware telemetry; MicroRCA [32] and TraceRCA [33] localize root causes over service-dependency and trace graphs; DeepTraLog [34] and Eadro [35] combine traces, logs and metrics under graph-based deep models. This line is the closest methodological neighbour to our predictive pathway, and it consistently outperforms what a purely static analysis can achieve — because it observes the running system. That is precisely the boundary: every one of these approaches requires a deployed system emitting traces, logs or metrics, and therefore cannot answer a question posed at design or pull-request time. SaG occupies the pre-deployment complement, and accepts a correspondingly weaker evidential basis: simulated rather than observed failures, and topology rather than behaviour.

2.2. Static Code Analysis (SCA) vs. Static System Analysis (SSA)

Traditional **Static Code Analysis (SCA)** tools (e.g., SonarQube [15]) inspect source code Abstract Syntax Trees (ASTs) within individual services. They evaluate cyclomatic complexity [16], class cohesion, module coupling (e.g., Lack of Cohesion in Methods [LCOM], Coupling Between Objects [CBO]) [17, 18], and code duplication to flag internal code smells and defect-prone modules [36, 37, 38, 39]. However, SCA cannot observe runtime communication topology: it is blind to inter-service messaging channels, message broker queue saturation, and cross-host failure propagation.

To bridge this “Architecture–Code Gap,” **Static System Analysis (SSA)** extends static analysis from single-service source code to the global system architecture. By modeling distributed applications, message topics, brokers, execution nodes, and shared libraries as a connected multigraph, SSA propagates code-level quality metrics across architectural dependencies. This allows engineering teams to detect structural anti-patterns [40, 41] and architectural technical debt [42] early during continuous integration (CI/CD) [43, 44], before defective topologies enter production.

2.3. Software Quality Models and Multi-Criteria Evaluation

Software product quality is standardized by the **ISO/IEC 25010:2023** product quality model [9] and the **ISO/IEC 25019:2023** Quality-in-Use model [10]. ISO/IEC 25010:2023 defines three closely intertwined characteristics critical to modern distributed systems:

- **Reliability:** The degree to which a system performs specified functions under stated conditions, comprising Faultlessness, Availability, Fault Tolerance and Recoverability.
- **Maintainability:** The degree of effectiveness and efficiency with which software can be modified, comprising Modularity, Reusability, Analysability, Modifiability and Testability.
- **Performance Efficiency:** Performance relative to resource consumption under stated conditions, comprising Time Behavior (latency, response time), Resource Utilization (CPU, memory, bandwidth), and Capacity.

SaG operationalizes a strict subset of these: Availability and Fault Tolerance under Reliability, and Modularity, Modifiability and Analysability under Maintainability (Section 5.1). Faultlessness, Recoverability, Reusability and Testability are not derivable from deployment topology alone and are outside the scope of this work.

Software engineering measurement explicitly distinguishes between *internal quality* (measured on static artifacts at rest) and *external quality* (measured on executing software systems) [45, 46]. In distributed architectures, architectural debt (such as over-centralized message topics or unreplicated brokers) degrades internal quality and precipitates severe external performance bottlenecks, queue congestion, and outages.

Aggregating multi-attribute structural metrics into an auditable quality score constitutes a classic Multi-Criteria Decision Making (MCDM) problem. The **Analytic Hierarchy Process (AHP)** [47] delivers a structured pairwise-comparison method with an explicit Consistency Ratio ($CR \leq 0.10$) to ensure mathematical soundness in weighting models. This study applies AHP to construct an audited, explainable Reliability–Maintainability (RM) quality baseline, in conjunction with learned graph models.

2.4. Graph Representation Learning and Explainable AI

Network science provides established centrality metrics to identify critical nodes, such as degree, closeness, betweenness centrality [20, 22], articulation points, and PageRank [21, 23]. Foundational studies on network robustness [48], cascading overloads [49], and interdependent networks [50] model how disruptions propagate across connected systems. However, standard network metrics suffer from two major limitations when applied to software architectures:

1. **Dimensional Collapse:** A single centrality scalar cannot distinguish *why* a component is critical—for instance, whether it is a single point of failure (SPOF), an error-propagating cascade hub, or an over-shared library.
2. **Semantic Collapse:** Standard metrics treat all nodes and edges identically. They conflate fundamentally different architectural entities, such as an asynchronous message topic, a shared library, and a physical execution host.

To overcome the limits of hand-engineered metrics, recent research has applied machine learning to network vulnerability (e.g., FINDER [51], DrBC [52], and PowerGraph [53]). However, most available models rely on **homogeneous message passing** (GCN [54], GraphSAGE [55], GAT [56]), which averages signals across all connections indiscriminately. Because distributed software architectures are inherently **heterogeneous** (comprising distinct entity types and relationship rules), homogeneous models blur critical architectural boundaries and fail to generalize out-of-distribution.

Heterogeneous Graph Neural Networks (RGCN [57], HAN [58], HGT [59], MAGNN [60]) resolve this by employing relation-specific transformations. We build upon the **Heterogeneous Graph Transformer (HGT)** architecture [59] to preserve typed relational semantics when forecasting cascading failure blast radii and performance degradation.

Explainable AI (XAI) vs. The Black-Box Barrier. A critical hurdle in applying modern AI to software engineering is the **black-box barrier**: deep neural models output risk scores or continuous embeddings without explaining underlying structural causality. In production software engineering, uninterpretable risk rankings hinder actionable decision-making: developers and SREs cannot determine whether to replicate a host, configure circuit breakers, or refactor shared libraries.

Existing GNN explanation techniques, such as GNNExplainer [61] and PGExplainer [62], identify influential subgraphs through edge masking or parameterized learning. Although useful, these methods explain the model using internal latent representations rather than standardized software engineering concepts. SaG resolves this limitation through a decoupled dual-pathway design: the predictive HGT pathway reveals typed mutual-attention distributions indicating *which* architectural relations propagated the cascade (Section 7.4), while the deterministic explanation layer attributes fragility to standardized ISO/IEC quality sub-characteristics (Section 5), translating raw predictions into actionable, cost-effective remediations.

3. The Software-as-a-Graph (SaG) Architectural Model

This section formalizes the Software-as-a-Graph multigraph representation (Section 3.1), the QoS-aware weighting and logical dependency derivation rules (Section 3.2), the dual graph views (Section 3.3), and the typed node feature encodings (Section 3.4).

3.1. Formal Multigraph Definition

A complex distributed software system is formally modeled as a typed, weighted, directed multigraph:

$$\mathcal{G} = (V, E, \tau_V, \tau_E, w_V, w_E) \quad (1)$$

where:

- V is the set of system entities, partitioned into five disjoint entity types:

$$V = V_{\text{app}} \cup V_{\text{broker}} \cup V_{\text{topic}} \cup V_{\text{node}} \cup V_{\text{lib}} \quad (2)$$

- E is the set of directed edges connecting entities.
- $\tau_V : V \rightarrow \mathcal{T}_V$ and $\tau_E : E \rightarrow \mathcal{T}_E$ are typing functions assigning node and edge categories.
- $w_V : V \rightarrow [0, 1]$ and $w_E : E \rightarrow [0, 1]$ are weighting functions representing entity criticality and connection strength.

Table 1 summarizes the five entity types and six structural edge types formalized in the SaG model, along with their semantics and representative concrete distributed systems implementations.

Table 1: Entity and structural edge types in the SaG model.

Entity Type ($\mathcal{T}_V$)	Architectural Role	Concrete System Examples
Application (V_{app})	Autonomous process producing/consuming messages	ROS 2 node, Kafka microservice, MQTT client
Broker (V_{broker})	Message routing and queuing intermediary	RabbitMQ exchange, Mosquitto, EMQX broker
Topic (V_{topic})	Named logical communication channel	<code>/sensor/lidar</code> , <code>orders.payment.completed</code>
Node (V_{node})	Physical host or virtualized execution environment	Bare-metal server, Kubernetes worker, Cloud VM
Library (V_{lib})	Shared software package or runtime dependency	<code>librdkafka</code> , OpenCV, Protobuf runtime
Structural Edge ($\mathcal{T}_E$)	Direction	Semantic Meaning
PUBLISHES_TO	App/Library $\rightarrow$ Topic	Component publishes messages to topic
SUBSCRIBES_TO	App/Library $\rightarrow$ Topic	Component consumes messages from topic
ROUTES	Broker $\rightarrow$ Topic	Broker manages and routes topic traffic
RUNS_ON	App/Broker $\rightarrow$ Node	Process is hosted on physical/virtual host
CONNECTS_TO	Node $\rightarrow$ Node	Physical network link between hosts
USES	App $\rightarrow$ Library	Application links to shared library dependency

Application and Library entities additionally incorporate static code metrics computed via Static Code Analysis (SCA) tools (`cm_*` attributes: lines of code, cyclomatic complexity, coupling between objects, LCOM), linking code-level fragility directly to topological analysis.

3.2. QoS-Aware Weights and Logical Dependency Derivation

In distributed middleware, communication links vary in coupling strength based on their Quality-of-Service (QoS) contracts. For instance, a **RELIABLE** topic with **TRANSIENT_LOCAL** durability binds communicating services substantially more tightly than a **BEST_EFFORT** telemetry stream.

Each topic t carries an intrinsic criticality weight $w(t) \in [0, 1]$ combining its declared QoS semantics with two runtime-stress modulators: payload size and publication frequency:

$$w(t) = \beta \cdot \text{QoS}(t) + \alpha \cdot \text{SizeNorm}(t) + \psi \cdot \text{FreqNorm}(t), \quad (\beta, \alpha, \psi) = (0.75, 0.15, 0.10) \quad (3)$$

where the QoS term is an AHP-weighted aggregate of the declared contract:

$$\text{QoS}(t) = w_{\text{rel}} \cdot q_{\text{rel}} + w_{\text{dur}} \cdot q_{\text{dur}} + w_{\text{prio}} \cdot q_{\text{prio}}, \quad (w_{\text{rel}}, w_{\text{dur}}, w_{\text{prio}}) = (0.24, 0.62, 0.14) \quad (4)$$

Here, $q_{\text{rel}}, q_{\text{dur}}, q_{\text{prio}} \in [0, 1]$ represent normalized reliability, durability, and transport-priority scores. Durability dominates because it governs whether data persists across restarts and network partitions. Reliability and transport priority both govern in-flight delivery quality, with reliability receiving higher weight because unconditional delivery guarantees precede message scheduling. The sub-weight vector is the geometric-mean priority vector of an independently stated Saaty pairwise-comparison matrix with a small, non-zero consistency ratio ($CR \approx 0.016 \leq 0.10$). The modulators are logarithmically compressed: $\text{SizeNorm}(t) = \log_2(1 + \text{bytes})/20$ (a 1 MiB design envelope, representing the practical DDS sample ceiling before RTPS fragmentation dominates) and $\text{FreqNorm}(t) = \log_{10}(1 + \text{Hz})/3$. The weight $w(t)$ is clamped to $[0.01, 1]$ ensuring that best-effort edges remain visible to graph traversals. Every structural communication edge incident on t (**PUBLISHES_TO**, **SUBSCRIBES_TO**, **ROUTES**) inherits $w_E(e) = w(t)$ along with the topic's QoS vector.

The outer split (β, α, ψ) is a declared convex combination whose sensitivity is evaluated in Section 7.4.2.

3.2.1. Logical Dependency Projection (**DEPENDS_ON**)

Structural edges capture explicit deployment connections but omit implicit runtime dependencies. For example, a subscriber depends upon a publisher, yet no direct edge connects them in pub-sub architectures. We therefore derive a single unified semantic relation, **DEPENDS_ON**, directed from *dependent* to *dependency* (“if target fails, source is impacted”), according to the six projection rules detailed in Table 2:

Table 2: The six **DEPENDS_ON** logical dependency projection rules.

Rule	Dependency Category	Structural Pattern (Dependent $\rightarrow$ Dependency)	Derived Weight (w)
1	app_to_app	Subscriber $\rightarrow$ Publisher (via shared Topic, incl. transitive USES)	$1 - \prod_{t \in T} (1 - w(t))$
2	app_to_broker	Publisher/Subscriber $\rightarrow$ Broker routing its topics	$1 - \prod_{t \in T} (1 - w(t))$
3	node_to_node	Host $\rightarrow$ Host (lifted from inter-host app dependencies)	Lifted max w
4	node_to_broker	Host $\rightarrow$ Broker (lifted from hosted app dependencies)	Lifted max w
5	app_to_lib	Application $\rightarrow$ Shared Library it USES	$H(w_V(\text{app}), w_V(\text{lib}))$
6	broker_to_broker	Broker $\leftrightarrow$ Broker (shared physical fault-domain colocation, symmetric)	$w_V(\text{node})$

Rules 1 and 2 aggregate the set of topics T connecting a component pair using a probabilistic union rather than a maximum [63, 64, 65]. This guarantees that additional parallel failure vectors increase coupling monotonically while keeping $w \in (0, 1]$. Rule 5 applies the harmonic mean $H(x, y) = 2xy/(x + y)$ [66] to combine the consuming Application's and the shared Library's vertex weights, balancing caller and dependency criticality. Rules 3 and 4 assign the maximum weight among component-level dependencies crossing the host boundary.

3.2.2. Sequential Cascades vs. Simultaneous Blasts

A foundational principle of the SaG model is distinguishing between two fundamentally different degradation modes:

- **Sequential Cascade (Rule 1):** When an application publisher fails, downstream subscribers suffer message starvation. The failure propagates hop by hop through message queues and topic buffers.

- **Simultaneous Blast (Rule 5):** When a shared software library or execution node crashes, all consuming applications and colocated brokers fail *instantaneously* in a single shared-fate event.

Preserving architectural entity types and relation-specific projection rules enables SaG to model both mechanisms, whereas untyped homogeneous graphs collapse them into indistinguishable edges.

Rule 6 is intentionally the only symmetric projection rule. It does not imply that one broker functionally depends on another, but rather that two brokers colocated on the same host share that host’s physical failure domain. This follows the same simultaneous-blast principle as Rule 5, which is why the derived weight equals the shared Node’s weight and the relation is bidirectional. In production middleware deployments, colocated brokers compete for host resources (CPU cores, page cache, file descriptors, and NIC bandwidth); a host outage takes down all colocated instances simultaneously. Operational best practices for Kafka, RabbitMQ, and EMQX recommend distributing brokers across fault domains. Rule 6 does not model directional intra-cluster broker coupling (e.g., partition replication, controller quorum election, federation, or shovel links), which do not require physical colocation; extending the schema to capture these interactions is reserved for future work. In our evaluation corpus, Rule 6 applies in four of the eight cached scenarios (the seven synthetic topologies of Table 4 plus the ATM case study of Section 6.1) and contributes only 12 directed edges in total. Because the simulation oracles operate strictly on $G_{\text{structural}}$ (Section 4.4), Rule 6 has zero influence on ground-truth failure labels.

3.3. Dual Graph Views and Architectural Layers

The SaG framework maintains two distinct representations of the system:

1. **Structural Graph ($G_{\text{structural}}$):** The raw deployment graph containing physical and structural relations (such as PUBLISHES_TO, ROUTES, RUNS_ON, and USES). Discrete-event simulators consume this view exclusively to execute unbiased failure injections (Section 4.3).
2. **Analysis Graph (G_{analysis}):** The projected graph containing derived DEPENDS_ON edges annotated with QoS weights and ingested SCA code metrics. All GNN feature representations, graph embeddings, and analytical metrics are computed on G_{analysis} .

Figure 2 illustrates this duality on a running example, contrasting the raw structural graph against the derived DEPENDS_ON projection.

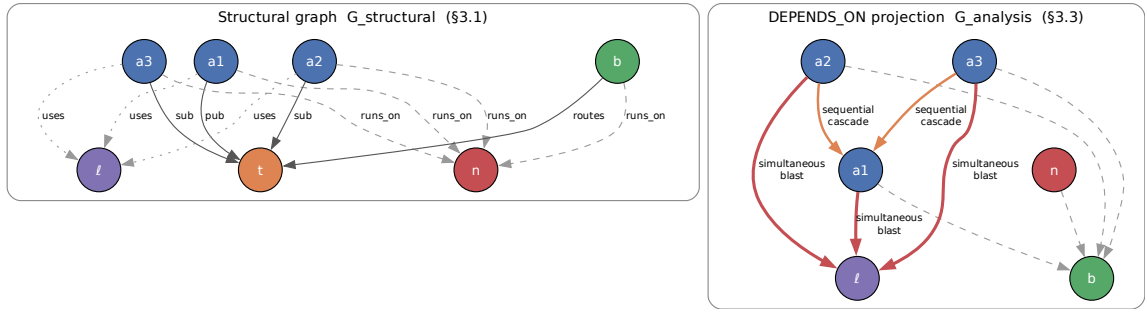


Figure 2: Running example: the raw structural graph (left) and the DEPENDS_ON projection derived from it (right). The projection makes implicit runtime dependencies explicit—a subscriber depends on the publishers of its topics even though no structural edge joins them—while the simulators continue to operate on the structural view alone.

G_{analysis} is further structured into four analytical layers (Application, Middleware, Infrastructure, and Global System), enabling evaluation of criticality at subsystem levels, consistent with hierarchical frameworks such as MIL-STD-498 [67].

3.4. Typed Node Feature Encoding

Both pathways read the same typed node properties from G_{analysis} : the predictive pathway (Section 4) projects them per entity type before heterogeneous message passing, and the explanation layer (Section 5) aggregates them into its quality profile. All five entity types share indices 0–17, a common block of topological metrics (in/out degree, betweenness, closeness, reverse PageRank, clustering coefficient, articulation score, bridge load) produced by the deterministic analysis stage whose cost is characterized in Section 7.6. Type-specific features extend that block:

- **Application (23 dims):** indices 18–22 add source-code metrics from SCA — lines of code, cyclomatic complexity, Martin’s instability $I_{\text{code}} = C_e/(C_a + C_e)$ [68], Lack of Cohesion in Methods, and the composite Code Quality Penalty (CQP).
- **Library (25 dims):** the Application block plus two library-specific blast-radius drivers (23–24): the normalized size of the transitive reverse-USES closure, and the normalized count of distinct subscribers reachable from topics published within that closure.
- **Broker (19 dims):** index 18 is normalized queue buffer capacity.
- **Topic (22 dims):** indices 18–21 are publisher count, subscriber count, log message frequency $\log(1 + \text{freq})$, and ordinal QoS criticality.
- **Infrastructure Node (20 dims):** indices 18–19 are normalized CPU core allocation and physical memory.

4. Graph Learning for Failure-Impact Prediction

Cascading failure impact in distributed software systems is inherently non-linear, multi-hop, and relation-dependent. Outages propagate not merely based on neighbor count, but through architectural relations and dependencies extending multiple hops beyond the initial fault. Whether a closed-form combination of standard centrality metrics can capture these compound dynamics is an empirical question rather than a settled one. The primary predictive pathway of Section 1.2 therefore employs a learned graph model, and Section 7.2 evaluates it against exactly such a closed-form baseline — which, on out-of-distribution ranking, it does not significantly surpass.

This section details the Heterogeneous Graph Transformer (HGT) architecture and its typed edge encodings (Section 4.1), the multi-task prediction heads and dimension-masked loss formulation (Section 4.2), the ground-truth simulation oracles (Section 4.3), and the input-label independence guarantee that prevents data leakage (Section 4.4).

4.1. Heterogeneous Graph Transformer Architecture

Because distributed systems comprise heterogeneous entity types (Applications, Libraries, Brokers, Topics, Infrastructure Nodes) and diverse interaction semantics (PUBLISHES_TO, SUBSCRIBES_TO, ROUTES, RUNS_ON, CONNECTS_TO, USES, DEPENDS_ON), we employ a three-layer **Heterogeneous Graph Transformer (HGT)** architecture [59], implemented within PyTorch Geometric [69], with hidden dimension $D = 64$ and $H = 4$ attention heads. This architecture ensures that typed relations, rather than simple adjacency, govern failure-impact forecasting.

4.1.1. Continuous-Categorical Edge Feature Encoding (16-D)

To capture continuous QoS constraints and channel semantics, SaG encodes each directed edge $e = (u, v)$ as a 16-dimensional continuous-categorical vector $e_{uv} \in \mathbb{R}^{16}$. Index 0 is the scalar coupling weight $w_E(e) \in (0, 1]$ of Section 3.2; index 1 is the normalized count of simple paths through e in G_{analysis} ; indices 2–8 one-hot encode the seven structural and derived relations (PUBLISHES_TO, SUBSCRIBES_TO, ROUTES, RUNS_ON, CONNECTS_TO, USES, DEPENDS_ON). Indices 9–15 carry the seven explicit QoS dimensions, active on PUBLISHES_TO and SUBSCRIBES_TO edges and zeroed elsewhere: reliability (0 best-effort, 1 reliable); durability (0 volatile,

0.5 transient-local, 0.6 transient, 1 persistent); message priority (0, 0.33, 0.66, 1); a deadline-active flag; $\log_{10}(1 + \text{deadline_ns}/10^6)$; $\log_{10}(1 + \text{max_blocking_ms})$; and a heterogeneity flag set when the edge’s QoS triple departs from its scenario’s modal profile.

An edge projection module maps e_{uv} into the hidden space: $e'_{uv} = W_{\text{edge}}e_{uv}$. Prior to relational attention computation, this projection vector is incorporated directly into the target node representation: $\tilde{h}_v = h_v + e'_{uv}$.

4.1.2. Type-Specific Projection and Heterogeneous Message Passing

For each source node u and target node v connected by meta-relation $\tau(e) = (\tau(u), \phi(e), \tau(v))$:

1. **Type-Specific Projection:** Node feature vectors x_v (of dimension 19–25 depending on entity type $\tau(v)$) are mapped into the shared D -dimensional hidden space:

$$h_v^{(0)} = \text{LayerNorm}(\text{GELU}(W_{\tau(v)}x_v)) \quad (5)$$

2. **Relational Mutual Attention:** Type-parameterized Query (Q), Key (K), and Value (V) projections calculate relation-specific attention across H heads:

$$\text{Attn}(u, e, v) = \text{Softmax}_{\forall u \in \mathcal{N}(v)} \left(\frac{K(u)W_{\text{att}, \phi(e)}Q(\tilde{h}_v)^\top}{\sqrt{D/H}} \right) \quad (6)$$

$$\text{Msg}(u, e, v) = V(u)W_{\text{msg}, \phi(e)} \quad (7)$$

3. **Bidirectional Message Passing:** To capture downstream consumer starvation and upstream back-pressure simultaneously, message passing is executed over both forward and transposed relation views (G_{analysis} and $G_{\text{analysis}}^\top$).
4. **Residual Aggregation and Layer Normalization:** Target node representations are updated across layers $l \in \{1, \dots, L\}$ via residual connections and layer normalization:

$$h_v^{(l)} = \text{LayerNorm} \left(h_v^{(l-1)} + \text{Dropout} \left(\sum_{u \in \mathcal{N}(v)} \text{Attn}(u, e, v) \cdot \text{Msg}(u, e, v) \right) \right) \quad (8)$$

Training Protocol and Optimization Hyperparameters. Models are optimized end-to-end using AdamW with initial learning rate $\eta = 3 \times 10^{-4}$, weight decay 10^{-4} , and dropout probability $p = 0.10$ applied post-attention. Learning rates follow a cosine annealing schedule with warm restarts (CosineAnnealingWarmRestarts, $T_0 = 75$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 3 \times 10^{-6}$). Training executes for a maximum of 300 epochs with early stopping governed by a patience of 30 epochs monitored on validation loss over labeled nodes. Inductive subgraphs are processed per scenario using full-graph inductive packing without mini-batch subsampling, with validation masks isolating held-out nodes to prevent information leakage across data splits. Five independent random seeds {42, 123, 456, 789, 2024} are evaluated across all runs, redrawing both partition masks and initializations. *Selection protocol:* the architectural hyperparameters ($D = 64$, $H = 4$, dropout, learning rate and schedule) and the loss coefficients of Equation (9) were fixed a priori from values conventional for HGT [59] and were not tuned against any evaluation reported in this paper; no search over them was performed, on either the in-distribution test split or the LOSO folds. This avoids selection leakage, at the cost of leaving open whether either family is reported near its own optimum — a comparison between untuned configurations, which we state rather than treat as a like-for-like optimum comparison.

4.2. Multi-Task Prediction Heads and Dimension Masking

From the final node embeddings $h_v^{(L)}$, SaG utilizes specialized multi-task prediction heads:

- **Reliability Head:** $\hat{R}(v) = \sigma(\text{MLP}_R(h_v)) \in [0, 1]$
- **Maintainability Head:** $\hat{M}(v) = \sigma(\text{MLP}_M(h_v)) \in [0, 1]$
- **Composite Failure Impact Head:** $\hat{I}^*(v) = \sigma(\text{MLP}_C(h_v \parallel \hat{R}(v) \parallel \hat{M}(v))) \in [0, 1]$
- **Relationship Criticality Head:** $\hat{Q}(u, v) = \sigma(\text{TypedEdgeEncoder}_{\phi(e)}(h_u, h_v, e_{uv})) \in [0, 1]$

4.2.1. Dimension-Masked Loss Formulation

The combined optimization objective integrates regression accuracy, multi-task dimension learning, ranking fidelity, pairwise ordering, and edge prediction:

$$\mathcal{L} = \mathcal{L}_{\text{composite}} + 0.5 \cdot \mathcal{L}_{\text{dimension}} + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}} + 0.3 \cdot \mathcal{L}_{\text{edge}} + \lambda_{\text{RM}} \cdot \mathcal{L}_{\text{consistency}} \quad (9)$$

where $I^*(v)$ is the simulated cascade impact defined by the primary oracle (Section 4.3), $\mathcal{L}_{\text{composite}} = \text{MSE}(\hat{I}^*(v), I^*(v))$, $\mathcal{L}_{\text{rank}}$ is the ListMLE ranking loss [70], $\mathcal{L}_{\text{pairwise}}$ is margin-ranking loss, and $\mathcal{L}_{\text{consistency}} = \text{MSE}([\hat{R}(v), \hat{M}(v)]_{v \in \text{unlabeled}}, [R_{\text{RM}}(v), M_{\text{RM}}(v)]_{v \in \text{unlabeled}})$ regresses predicted heads toward the diagnostic pathway’s baseline (Section 5) on unlabeled nodes. Headline results use $\lambda_{\text{RM}} = 0$, guaranteeing that the predictive and explanatory pathways remain strictly independent.

Dimension Masking: Because dynamic cascade simulation ($I^*(v)$ via **FaultInjector**) observes runtime failure reachability rather than source-code maintainability, maintainability ground truth is unobserved during dynamic simulation. A separate change-propagation oracle $I_M(v)$ evaluates static structural change ripple at the Validate stage, but is never used as a training label to avoid circular supervision. We introduce a boolean dimension mask $m = [m_R, m_M] = [1, 0]$:

$$\mathcal{L}_{\text{dimension}} = \frac{1}{\sum_d m_d} \sum_{d \in \{R, M\}} m_d \cdot \text{MSE}(\hat{d}(v), d^*(v)) \quad (10)$$

This mask ensures the unobserved maintainability head is not artificially penalized or driven toward zero during backpropagation.

4.2.2. Domain-Rewighted Criticality (Q_{domain})

To ground predictions in ISO/IEC 25019 Context of Use ($\vec{\omega} = [q_R, q_M]^\top$), the composite score can be evaluated as:

$$Q_{\text{domain}}(v) = q_R \cdot \hat{R}(v) + q_M \cdot M_{\text{static}}(v) \quad (11)$$

where $M_{\text{static}}(v)$ is obtained directly from the structural analyzer’s maintainability baseline, combining learned dynamic reliability with static source-code maintainability. Because maintainability is unobserved in dynamic simulation ($m = [1, 0]$), headline results report $\hat{I}^*(v)$ directly, while domain reweighting sensitivity is evaluated against the static RM baseline in Section 7.4.

4.3. Ground-Truth Simulation Oracles

To evaluate predictive accuracy prior to deployment without relying on production runtime telemetry, SaG executes discrete-event failure simulations over the raw structural multigraph $G_{\text{structural}}$. We establish a formal taxonomy of four component-level oracles and one relationship-level oracle:

- **Cascade Reachability Oracle ($I^*(v)$):** Implemented via **FaultInjector**, it crashes component v , propagates outages across dependent topics, brokers and links by breadth-first traversal, and returns the mean fractional feed loss over the subscriber population, each subscriber contributing the unweighted mean loss of the topics it subscribes to. The denominator is the full set of subscribers in the intact graph: subscribers that themselves fail during the cascade are retained in the average rather than excluded from it. A topic’s feed loss is the fraction of its publishers that have failed (for a topic with no publisher, the fraction of its failed routers), scaled by a QoS ladder ($\times 1.2$ **RELIABLE**, $\times 1.15$ high/urgent priority, $\times 1.05$ medium) and clamped to $[0, 1]$. The implementation admits a per-publisher rate weighting, but publication rates are declared per topic rather than per publisher throughout our corpus, so every publisher of a topic carries the same rate and the rate-weighted form reduces exactly to this publisher fraction. This is the **primary continuous target label** throughout.
- **Multi-Metric Composite Oracle ($I_{\text{comp}}(v)$):** Implemented via **FailureSimulator**, this oracle evaluates a multi-faceted failure impact vector:

$$I_{\text{comp}}(v) = 0.35 \cdot \Delta \text{Reachability} + 0.25 \cdot \Delta \text{Fragmentation} + 0.25 \cdot \Delta \text{Throughput} + 0.15 \cdot \Delta \text{FlowDisruption} \quad (12)$$

where each term is weighted by operational severity $s(t) = w(t) \cdot \text{rate}(t)$. $I_{\text{comp}}(v)$ serves as the canonical oracle for architectural quality gate verification.

Reading the rate modulator. $\text{rate}(t)$ is the topic’s observed message rate, read from a runtime telemetry profile when one is attached and defaulting to unity otherwise. No such profile is attached in any evaluation reported in this paper, which draws its ground truth from structural models rather than production observation; $s(t) = w(t)$ throughout Section 7, and the rate channel should be read as an interface for deployment-time calibration rather than as an active term in our results. Publication frequency and payload size nonetheless reach $s(t)$, through the FreqNorm and SizeNorm modulators internal to $w(t)$ (Section 3.2). The severity term is therefore a function of declared QoS semantics and objective physical topic attributes only.

- **Dynamic Queue-Flow Oracle ($I_{\text{dyn}}(v)$):** Implemented via `MessageFlowSimulator` using the SimPy framework [71], this oracle simulates message emission rates, stochastic network latencies, broker buffer saturation, and queue drops under fault injection. It extracts the drop in delivered message rate suffered by surviving consumers.

- **Change-Propagation Oracle ($I_M(v)$):** Implemented via `ChangePropagationSimulator`, this oracle executes a deterministic breadth-first traversal of the reversed dependency graph to quantify maintenance change impact:

$$I_M(v) = 0.45 \cdot \text{ChangeReach}(v) + 0.35 \cdot \text{WeightedChangeImpact}(v) + 0.20 \cdot \text{NormalizedChangeDepth}(v) \quad (13)$$

- **Relationship (Edge) Removal Oracle ($I_{\text{edge}}(u, v)$):** Evaluates the systemic impact of severing an individual dependency or communication channel while keeping endpoint components operational. Writing $\bar{I}_{\text{comp}}(G) = |V|^{-1} \sum_{v \in V} I_{\text{comp}}(v; G)$ for the mean composite impact over a graph G :

$$I_{\text{edge}}(u, v) = \bar{I}_{\text{comp}}(G \setminus \{(u, v)\}) - \bar{I}_{\text{comp}}(G) \quad (14)$$

Withholding declared topic criticality from labels. `FailureSimulator` can blend a declared `Topic.criticality` into its severity term; this is disabled, because that field is a GNN input feature (`topic_qos_criticality_ord`) and consuming it would score the predictor against a transformation of its own input.

Primary Oracle Declaration and Role Assignment. Because the three reliability-facing oracles (I^* , I_{comp} , I_{dyn}) measure distinct operational constructs, we designate $I^*(v)$ (**FaultInjector**) as the **primary oracle** for all predictive ranking results (Tables 6–8, RQ1–RQ3). $I_{\text{comp}}(v)$ is reserved for Validate-stage quality gates, $I_{\text{dyn}}(v)$ serves as an independent convergent-validity probe, and $I_M(v)$ serves as a structural maintainability reference.

Cross-Oracle Convergent Validity and Critical-Set Bounds. Measured across seven benchmark scenarios and five random seeds on the Application population, the mean Spearman rank correlation is $\rho = 0.907$ for (I_{dyn}, I^*) , $\rho = 0.425$ for (I_{comp}, I^*) , and $\rho = 0.427$ for $(I_{\text{comp}}, I_{\text{dyn}})$. The strong rank agreement ($\rho = 0.907$) between the behavioral queue-flow oracle and the topological cascade injector provides independent convergent evidence across distinct simulation paradigms. However, agreement on the top- K critical set ($K = 0.2n$) is more conservative (mean Jaccard overlap of 0.49 for the strongest pair and 0.24–0.28 for the two I_{comp} pairs, vs. 0.111 expected by chance), highlighting the intrinsic sensitivity of discrete thresholding in non-linear cascades. Consequently, results established against one oracle are never transferred to another; every evaluation metric explicitly references its underlying simulation oracle.

4.4. Input-Label Independence Guarantee

To eliminate data leakage and ensure rigorous evaluation, SaG enforces strict architectural separation between inputs and labels:

- **Feature Space:** Constructed exclusively from G_{analysis} using static structural topology, static code analysis (SCA) metrics, and declared QoS contracts.

- **Label Space:** Evaluated exclusively on raw $G_{\text{structural}}$ through independent simulation oracles (`FaultInjector`, `FailureSimulator`, `MessageFlowSimulator`).

No simulation outputs, failure trace histories, or dynamic execution telemetry are ever exposed as input features to the GNN or the explanation layer.

5. The Explanation Layer: Standards-Grounded Criticality Attribution

The predictor of Section 4 answers *where* to act. It does not answer *what to do*, and neither does the oracle that scores it — both return impact, not cause attributed in standardized quality terms. A component may be critical because it is a single point of failure, because it propagates errors widely, or because it is a high-coupling maintenance bottleneck. These diagnoses call for distinct architectural repairs: replicating the host or broker, decoupling the topic, or refactoring the module. This section presents the layer supplying that diagnosis. SaG decomposes component and relationship criticality into a standards-grounded quality profile, computed over the same typed node features (Section 3.4) but sharing no parameters with the predictor, and applied to whatever the predictor flagged — triage rather than data flow (Figure 1).

5.1. Grounding in ISO/IEC Standards

In accordance with **ISO/IEC 25010:2023** (Product Quality Model) [9], **ISO/IEC 25019:2023** (Quality-in-Use) [10], and **ISO/IEC 25022:2016** (Measurement of Quality-in-Use) [72], SaG formalizes two primary criticality constructs:

- **Component Criticality (D_1):** The degree to which the sudden failure, unexpected termination, or severe degradation of an individual component reduces the system’s capacity to deliver required services within its operational context of use.
- **Relationship Criticality (D_2):** The degree of systemic service degradation resulting from the severance, partitioning, or failure of a specific dependency or communication channel while both endpoint components remain operational.

Criticality is evaluated across two orthogonal quality characteristics: **Reliability (R)** and **Maintainability (M)**. The two are separated because they imply different repairs, not because they predict different runtime quantities: this layer attributes structural fragility and does not forecast latency, throughput or queue occupancy. Table 3 outlines this Reliability–Maintainability (RM) quality decomposition, mapping ISO/IEC sub-characteristics to architectural questions, underlying graph metrics, and targeted engineering remediations.

Table 3: The Reliability–Maintainability (RM) quality decomposition.

Dimension	Sub-Characteristic	Architectural Question	Underlying Graph Metrics	Role / Remediation
Reliability (R)	Fault Tolerance (FT)	How broadly does failure propagate?	Reverse PageRank on G^T , in-degree, cascade depth	Reliability Eng.: add redundancy, circuit breakers
	Availability (A)	Is this a single point of failure?	Directed articulation score (raw + QoS-weighted), bridge ratio, connectivity degradation	DevOps/SRE: replicate host/broker
Maintainability (M)	Modularity/Modifiability	How complex and coupled is this?	Betweenness, QoS-weighted out-degree, Code Penalty, coupling-risk imbalance, inverse clustering	Architect: refactor code, decouple

Coverage Scope: SaG focuses specifically on Reliability and Maintainability. Safety (which requires domain-specific hazard logs, such as ISO 26262 Automotive Safety Integrity Level [ASIL] ratings) and Security (which requires explicit threat models, such as STRIDE [Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege]) fall outside purely structural topology analysis and are reserved for domain-specific extensions.

5.2. Composite Quality Score Formulation

All raw topological and code metrics are rank-normalized to $[0, 1]$. Quality sub-characteristics are formulated hierarchically using the Analytic Hierarchy Process (AHP) [47]:

1. **Fault Tolerance** ($FT(v)$): Measures error cascade potential on the transpose graph G_{analysis}^T (where edges follow failure propagation from dependency to dependent):

$$FT(v) = 0.45 \cdot \text{RPR}(v) + 0.30 \cdot \text{Deg}_{\text{in}}(v) + 0.25 \cdot \text{CDPot}_{\text{enh}}(v) \quad (15)$$

where $\text{RPR}(v)$ is Reverse PageRank, $\text{Deg}_{\text{in}}(v)$ is normalized in-degree, and $\text{CDPot}_{\text{enh}}(v)$ is the enhanced Cascade Depth Potential term.

2. **Availability** ($A(v)$): Identifies structural single points of failure (SPOFs) across five terms — directed articulation severity, its QoS-weighted variant, edge-level irrecoverability, connectivity degradation, and the component’s own QoS weight:

$$A(v) = 0.2563 \cdot \text{AP}_c^{\text{dir}}(v) + 0.1998 \cdot \text{QSPOF}(v) + 0.1998 \cdot \text{BR}(v) + 0.2563 \cdot \text{CDI}(v) + 0.0878 \cdot w(v) \quad (16)$$

where $\text{AP}_c^{\text{dir}}(v)$ is Directed Articulation Point severity, $\text{QSPOF}(v)$ is QoS-weighted Single Point of Failure severity, $\text{BR}(v)$ is Bridge Ratio (edge-level irrecoverability), $\text{CDI}(v)$ is Connectivity Degradation Index, and $w(v)$ is the component’s intrinsic QoS weight.

3. **Reliability** ($R(v)$): Blends Fault Tolerance and Availability hierarchically:

$$R(v) = r_\alpha \cdot FT(v) + (1 - r_\alpha) \cdot A(v), \quad r_\alpha = 0.36 \quad (17)$$

The blend weight is written r_α throughout to distinguish it from the payload-size coefficient α of Equation (3); the two are unrelated parameters and are screened separately in Table 12. Intra-dimension pairwise comparison matrices are audited against Saaty’s consistency ratio and measure $CR = 0.001$ (Fault Tolerance), $CR = 0.001$ (Availability), and $CR = 0.000$ (Maintainability) — all well within the $CR \leq 0.10$ acceptability threshold. The shipped intra-dimension weights apply a $\lambda = 0.70$ shrinkage blend between the raw AHP-derived vector and a uniform prior (Section 7.4 reports ranking sensitivity to λ).

4. **Maintainability** ($M(v)$): Evaluates structural coupling combined with code-level static analysis across five terms — betweenness, QoS-weighted efferent coupling, the Code Quality Penalty, an afferent/efferent coupling-risk imbalance term, and inverse clustering:

$$M(v) = 0.35 \cdot \text{BT}(v) + 0.30 \cdot w_{\text{out}}(v) + 0.15 \cdot \text{CQP}(v) + 0.12 \cdot \text{CouplingRisk}_{\text{enh}}(v) + 0.08 \cdot (1 - \text{CC}(v)) \quad (18)$$

where $\text{BT}(v)$ is Betweenness Centrality, $w_{\text{out}}(v)$ is QoS-weighted efferent coupling (out-degree), $\text{CQP}(v)$ is the Code Quality Penalty, $\text{CouplingRisk}_{\text{enh}}(v)$ is an afferent/efferent coupling-risk imbalance term, and $\text{CC}(v)$ is the local Clustering Coefficient.

The baseline composite quality score $Q(v)$ combines both dimensions:

$$Q(v) = 0.80 \cdot R(v) + 0.20 \cdot M(v) \quad (19)$$

When evaluating under a specific ISO/IEC 25019 Context of Use vector $\vec{\omega} = [q_R, q_M]^T$, the score is reweighted dynamically:

$$Q_{\text{domain}}(v) = q_R \cdot R(v) + q_M \cdot M_{\text{static}}(v) \quad (20)$$

Components are categorized into adaptive criticality tiers using box-plot quartile thresholds:

- **CRITICAL:** $Q > Q_3 + 1.5 \cdot \text{IQR}$
- **HIGH:** $Q_3 < Q \leq Q_3 + 1.5 \cdot \text{IQR}$
- **MEDIUM:** $Q_1 < Q \leq Q_3$
- **MINIMAL:** $Q \leq Q_1$

This yields a directly actionable distinction: a component scoring high on A but low on FT is a single point of failure calling for horizontal replication, whereas one scoring high on FT is an error-cascade hub calling for circuit breakers, rate limiting and bulkhead isolation. Whether acting on these diagnoses improves any measured runtime property is not evaluated here (Section 8.4).

6. Experimental Setup

6.1. Datasets and System Corpus

The evaluation corpus comprises 1,896 components distributed across twelve distinct system architectures, as detailed in Table 4:

Table 4: Experimental evaluation corpus.

Dataset / Architecture	System Paradigm	$ V $	$ V_{\text{app}} $	Topics	Brokers	Hosts	Libs	$ E $
Autonomous Vehicle (AV)	ROS 2 Cyber-Physical	152	80	40	4	8	20	797
Enterprise Pub-Sub	Kafka Event Mesh	520	300	120	10	40	50	3,245
Financial Trading	Low-Latency Pub-Sub	124	60	35	5	6	18	580
Healthcare Integration	HL7/FHIR Event Mesh	98	50	25	3	8	12	400
Hub-and-Spoke Enterprise	Broker-Centric Messaging	139	70	30	2	12	25	797
IoT Smart City	MQTT Telemetry Mesh	326	200	80	6	30	10	1,322
Microservices Mesh	Cloud-Native Services	186	90	45	6	15	30	680
Autoware.universe [73]	Real-World ROS 2 Autoware	75	32	24	3	6	10	179
Cloud Microservices [74]	Real-World GCP Boutique	60	22	20	4	6	8	128
Train-Ticket [75]	Real-World Microservices	90	41	30	3	8	8	162
Home Assistant [76]	Real-World Smart Home	63	24	22	3	6	8	119
EdgeX Foundry [77]	Real-World Industrial IoT	63	22	24	3	6	8	112
Total		1,896						

Here, $|V|$ is the sum of all five entity-type counts per scenario, totaling exactly 2,387 components. $|E|$ counts every raw structural relationship instance recorded in the scenario specification (PUBLISHES_TO, SUBSCRIBES_TO, ROUTES, RUNS_ON, CONNECTS_TO, USES) — the native substrate that simulation oracles traverse — rather than the derived DEPENDS_ON projection constructed for GNN training.

Five real-world architectures were transcribed from authentic open-source repositories using dedicated architectural adapters. The eleven synthetic scenarios were produced by a parameterized topology generator: each is fully defined by a committed configuration specifying a random seed, per-entity-type counts, seven-number summaries (mean, median, standard deviation, minimum, maximum, Q_1 , Q_3) for application publish and subscribe fan-out, applications per host, library fan-in, topic payload size, and categorical distributions over the three QoS dimensions. Graph degree distributions and clustering emerge directly from these parameters rather than being synthetically forced. Table 5 reports the generative parameters governing each topology’s shape; complete configurations are included in the replication package.

Table 5: Generative parameters of the eleven synthetic evaluation scenarios. Counts, seed and fan-out figures are read directly from the committed configurations. The modal QoS column gives the most common reliability/durability/priority value and the range of topic shares carrying them, computed from the committed topology rather than the config’s declared QoS targets, which domain-driven assignment does not always realize (Section 6.1).

Scenario	Config	Seed	Counts	Pub	Sub	Modal QoS
Autonomous Vehicle (AV)	scenario_01_autonomous_vehicle	1001	80/40/4/8/20	2.5	5.0	RELIABLE/T
Enterprise Pub-Sub	scenario_07_enterprise_xlarge	7007	300/120/10/40/50	3.0	4.5	RELIABLE/T
Financial Trading	scenario_03_financial_trading	3003	60/35/5/6/18	4.0	6.0	RELIABLE/P
Healthcare Integration	scenario_04_healthcare	4004	50/25/3/8/12	2.5	3.0	RELIABLE/P
Hub-and-Spoke	scenario_05_hub_and_spoke	5005	70/30/2/12/25	2.0	7.0	RELIABLE/T
Industrial SCADA	scenario_19_industrial_scada	1902	140/70/4/25/15	1.1	1.0	RELIABLE/T
IoT Smart City	scenario_02_iiot_smart_city	2002	200/80/6/30/10	2.0	1.5	BEST_EFFORT
Logistics Fleet	scenario_21_logistics_fleet	2104	110/50/7/18/20	1.8	1.9	RELIABLE/T
Microservices Mesh	scenario_06_microservices	6006	90/45/6/15/30	1.5	2.0	RELIABLE/T
Real-Time Gaming	scenario_20_realtime_gaming	2003	75/38/5/12/28	2.6	2.8	RELIABLE/T
Telecom RAN	scenario_18_telecom_ran	1801	120/55/8/20/22	2.2	1.6	RELIABLE/T

QoS diversity across the corpus. Per-topic QoS profiles come from domain-keyed lookups that override the generic categorical distributions. Three domains (**hub-and-spoke**, **microservices**, **enterprise**) receive

an identical (reliability, durability, priority) triple across all topics and a fourth (**av**) collapses four of five entries to one profile, giving 85%–100% concentration at the modal triple against 51%–79% for the genuinely multi-valued domains (**finance**, **healthcare**, **iot**); see Table 5. The scalar coupling weight $w(t)$ therefore has standard deviation ≈ 0.02 in those four scenarios against 0.19–0.28 in the other three. This bounds what our QoS ablations can establish (Sections 7.4 and 8.3); we report it as a corpus property rather than forcing artificial variation.

Role of the ATM case study. An additional Automated Teller Machine network scenario serves as the twelfth LOSO fold and as the substrate for the attention analysis of Section 7.4. It is omitted from Table 4 because it was authored as an illustrative walkthrough and its specification lacks the seven-number summaries the other scenarios carry — so it participates in the headline LOSO result without being characterized alongside the rest of the corpus, which we note as a reproducibility limitation. Every LOSO evaluation therefore reports twelve folds: the eleven synthetic topologies of Table 4 plus ATM. The five real-world systems are never LOSO folds; they are withheld entirely and used only for the zero-shot transfer evaluation of Section 7.5.

6.1.1. Reproducibility of the Corpus

The benchmark corpus is designed to be fully regenerable rather than merely statically archived. Each dataset is deterministically generated from its configuration file via:

```
python cli/generate_graph.py batch -input-dir data/scenarios -output-dir <dir>
```

A companion manifest records the random seed, entity counts, git commit hash, and a SHA-256 cryptographic digest for each emitted topology. Continuous integration regression tests assert that every committed dataset regenerates *byte-identically* from its configuration and that all disk digests match the manifest. This guarantees that third parties can reproduce the exact graphs used in our experiments, rather than simply sampling from similar distributions.

6.2. Baselines and Evaluated Predictors

We evaluate four primary predictor configurations drawn from three families. Predictor names state the family and the substrate: an **-N** suffix marks a model trained on the *native* multigraph, its absence the derived Application–Library flow projection, and a **-QoS** suffix marks a configuration that consumes declared QoS contracts. *SaG* throughout denotes the framework, never an individual predictor.

1. **Heterogeneous graph learning (typed HGT). HGT-QoS** (proposed): relation-specific Heterogeneous Graph Transformer (Section 4) ingesting the complete native multigraph with 16-dimensional continuous-categorical edge features that encode middleware QoS contracts. Its ablation **HGT**, which masks those QoS dimensions, is reported in Section 7.4.1.
2. **Homogeneous graph learning (untyped GAT). GAT-N-QoS**: homogeneous Graph Attention Network [56] trained on the identical native multigraph substrate with per-type input projections, but untyped, single-relation message passing. Its edge channel carries the scalar QoS aggregate $w(e)$ — dimension 0 of the same 16-D encoding HGT-QoS consumes — rather than the per-dimension decomposition; no homogeneous architecture in our suite ingests the full 16-D vector. The HGT-QoS–GAT-N-QoS contrast therefore bounds the *joint* contribution of relational typing and per-dimension QoS encoding. Section 7.4.1 separates the second factor within the typed architecture, and the corresponding unweighted ablation is **GAT-N**.
3. **Structural baselines (training-free). Topo-QoS**: QoS-weighted topological centrality evaluated on the derived application flow projection.
4. **Structural baselines (training-free). Topo**: structural centrality combining unweighted betweenness centrality and articulation point scoring on the flow projection.

In addition, the out-of-distribution evaluation (Table 8) reports **RM** ($Q(v)$, the deterministic hierarchical quality attribution model of Section 5) as a diagnostic reference baseline. RM is not fitted to rank failure impact; its inclusion demonstrates how much learned relational prediction adds over static structural attribution (Section 1.2). Furthermore, deterministic RM scoring drives every sensitivity sweep in Section 7.4, where closed-form formulations isolate parameter effects from neural training stochasticity.

6.2.1. Evaluation Substrates and Parity Guarantee

Predictors in this study operate on matched substrates within their respective families:

- **Graph Learning Models (GAT-N-QoS, HGT-QoS):** Both learned neural predictors ingest the complete native typed multigraph across all five entity types in both in-distribution (Table 6) and out-of-distribution Leave-One-Scenario-Out (Table 8) evaluations — which is what the shared `-N` suffix records — so the comparison carries no multi-entity visibility confound. Node type reaches homogeneous GAT-N-QoS only through its per-type input projection layer, while message passing uses untyped GATConv with shared weights across all edges; in contrast, heterogeneous HGT-QoS employs relation-specific HGTCov weight matrices per edge triple alongside edge-type encodings. Substrate and node features are matched; the edge channel is not, since GAT-N-QoS consumes the scalar QoS aggregate $w(e)$ where HGT-QoS consumes all 16 dimensions (Section 6.2). Comparisons between the two therefore isolate relation-specific parameterization jointly with per-dimension QoS encoding, and we report the QoS factor separately in Section 7.4.1 rather than attributing the whole margin to typing.
- **Training-Free Structural Baselines (Topo, Topo-QoS):** Topological baselines are evaluated on the derived Application–Library `DEPENDS_ON` projection (Section 3.2). This projected substrate is necessary because in raw publish–subscribe multigraphs, Application nodes never route messages directly, resulting in near-zero betweenness and bridge ratios that yield degenerate, uninformative scores.
- **Ground-Truth Independence Guarantee:** Crucially, **no predictor in either group accesses $G_{\text{structural}}$** : that raw topology is strictly reserved as the substrate for ground-truth simulation oracles (Section 4.4), a guarantee formally verified by `tests/test_independence_guarantee.py`.

Regardless of substrate, all variants are scored on an identical, independently resolved Application node set (V_{app} , Section 6.3).

6.3. Evaluation Metrics and Protocols

Figure accessibility. All figures in this paper use the high-contrast, colorblind-safe Okabe–Ito palette together with distinct marker, hatching and node-shape encodings, so that every distinction carried by colour is also carried by form and remains legible in monochrome.

- **Ranking Precision:** Evaluated via Spearman rank correlation (ρ) and Kendall’s rank correlation (τ) between predicted component rankings and ground-truth simulated impact $I^*(v)$ from the primary oracle (Section 4.3).
- **Critical-Set Identification:** Measured via $F_1@K$, Precision@ K , and Recall@ K for top- K critical components, where $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$. Because predicted and ground-truth sets both contain exactly K elements, Precision, Recall, and F_1 coincide identically as the top- K set overlap.
- **Statistical Significance:** Assessed through paired Wilcoxon signed-rank tests [78] ($p < 0.05$) and non-parametric bootstrap 95% confidence intervals over folds ($B = 2,000$) [79, 80], reported for every out-of-distribution predictor in Table 8 and for the paired contrasts in Section 7.2. Given the sample size, we regard those intervals rather than the p -values as the primary evidence. The unit of analysis is the scenario (in-distribution, $n = 7$) or the fold (LOSO, $n = 12$). The in-distribution design sits at the floor of its own resolving power — the smallest attainable two-sided p is 0.0156 at $n = 7$, so only a near-unanimous sign pattern can register at all. The twelve-fold LOSO design is materially better resourced: its floor is 0.00049, and it tolerates as many as four lost folds while still reaching $\alpha = 0.05$, provided those losses are the smallest in magnitude. Where a LOSO comparison nonetheless fails to reach significance (Section 7.2), the cause is therefore the size of the losing folds rather than the size of the corpus, and enlarging the corpus would not remedy it. We report p -values uncorrected and treat them as exploratory. Under a family-wise correction no in-distribution comparison in Table 7 can reach $\alpha/6 = 0.0083$ regardless of the data, and the unanimous 8/8 LOSO results survive a six-comparison correction ($\alpha/6 = 0.0083$) but not a seven-comparison one ($\alpha/7 = 0.0071$). Directional consistency across folds, rather than any single p -value, is what carries the out-of-distribution claims.

6.3.1. Evaluation Population

Every predictor within a given evaluation table is scored on an identical node population, resolved strictly from scenario topology and simulation ground truth — never from any model’s predictions. Unless otherwise noted, this population is the **Application** set (V_{app}). This aligns with the framework’s primary objective (forecasting application-layer cascading failures) and ensures a fair common denominator across both typed and untyped predictors. Pooling node types into a single global ranking conflates distinct base rates and impact distributions, shifting the resulting rank correlation outside the envelope of per-type correlations (Section 7.4). We therefore report stratified, single-population metrics throughout and explicitly identify any pooled figures.

6.3.2. Evaluation Protocols

- **In-Distribution Evaluation:** 60% train / 20% validation / 20% test node splits over five seeds {42, 123, 456, 789, 2024}. Each split is a deterministic function of node identity *and* seed, so a seed redraws the partition as well as the initialisation. The consequence for Table 6 is that dispersion across seeds is split noise on 10–60 held-out nodes for every variant — including the training-free baselines, whose scores are otherwise deterministic — and additionally training noise for the learned predictors. Per-seed standard deviations are reported in that table for this reason.
- **Inductive Leave-One-Scenario-Out (LOSO):** Models are trained on eleven scenarios and evaluated zero-shot on the held-out twelfth, testing zero-shot generalization across distinct architectural domains. Three protocol properties of the harness bear on reproducibility, and all three are corrections to an earlier version whose consequences are quantified in Section 7.3.1. First, *every* learned variant receives the identical training set: one scenario is designated the *primary* graph and the remainder are passed as additional inductive graphs, for the homogeneous and heterogeneous architectures alike. Second, message-passing depth is fixed at three layers for every fold rather than derived from the primary graph’s size, so that model capacity does not vary with which scenario is held out. Third, early stopping and checkpoint selection are driven by a *held-out training scenario* — deterministically the median-sized inductive graph, excluded from the loss entirely — rather than by a within-scenario validation split of a training graph; selecting on a split of a graph the model is already fitting selects for in-distribution fit under a protocol whose purpose is distribution shift. The outer holdout participates in none of these choices. All three behaviours are fixed in the released harness and apply identically to every variant.
- **Real-World Architectural Transfer:** Evaluating models trained on synthetic corpora zero-shot on authentic open-source distributed systems without fine-tuning.

7. Results and Empirical Analysis

7.1. Artifact Provenance and Known Staleness

This section reports results measured across two states of the evaluation corpus, and we identify which is which before presenting any of them.

During preparation we found that the scenario generator never read the per-scenario `qos_stats` distributions declared in its configurations: every scenario declaring a domain — all of them — was routed through a name-pattern lookup instead, and the four newest domains, absent from that table, fell through to a single default row. Seven of the twelve scenarios consequently assigned one identical QoS triple to every topic. Repairing the branch eliminated the degeneracy (zero degenerate scenarios of twelve, 13–30 distinct triples each) and left the topology untouched: entity and relationship counts hold, and the pooled evaluation population remains 2,387.

The corpus was regenerated on that repair. Topology being unchanged does not make the results transferable, because $I^*(v)$ applies a QoS ladder to its severity term and $w(t)$ feeds every derived edge weight — so both labels and features move, and every number measured before the repair is superseded in principle even where it is unlikely to move much in practice.

Four artifacts postdate the regeneration and are current: the QoS corpus diagnostic (Section 7.4.1), the three-oracle agreement of Table 13, and the two real-world evaluations of Section 7.5.1. Everything else in this section predates it, including Table 6, the headline LOSO results of Table 8, the harness attribution of Table 9, the nested selection of Section 7.2.2, and the sensitivity sweeps of Tables 10, 11 and 12 with Figure 4. Their captions are marked accordingly. Regenerating them is mechanical — the ground-truth caches were rebuilt against the repaired corpus and the harnesses read them directly — and it is the work this manuscript owes before it can claim that every reported number was measured on the corpus it ships.

A second, independent gap affects Table 6 only. Its in-distribution protocol enumerates a scenario list that was fixed when the corpus held seven scenarios and never extended when four domains were added, so the in-distribution results cover seven of the eleven synthetic architectures while the LOSO results of Table 8 cover all twelve folds. The absent domains are Telecom RAN, Industrial SCADA, Real-Time Gaming and Logistics Fleet. We flag this rather than presenting Table 6 as corpus-wide, and note that the in-distribution typing margin of Section 7.3 (+0.088, 6/7) is measured on that reduced set.

We report all of this in the body rather than in a limitations paragraph because the replication package ships with the paper: a reader who regenerates the corpus and re-runs these harnesses will not reproduce the tables below, and the reason should not have to be inferred.

7.2. RQ1: Graph Learning vs. Structural Baselines

Table 6 presents the in-distribution held-out Spearman rank correlation (ρ) against simulated cascade impact $I^*(v)$ across all seven synthetic scenarios.

Table 6: In-distribution held-out Spearman rank correlation (ρ) against $I^*(v)$, reported as the mean over five seeds $\pm$ the standard deviation across those seeds; n is the held-out Application count. All four learned variants are evaluated on the identical native multigraph, differing only in typing (GAT-N vs. HGT) and edge channel (none vs. scalar $w(e)$ vs. all 16 dimensions, Section 6.2). Topological baselines operate on the flow projection. Each seed redraws the 60/20/20 split as well as the initialisation (Section 6.3). Read alongside the paired tests of Table 7, which use the scenario as the unit of analysis. Measured before the corpus repair of Section 7.1. Covers seven of the eleven synthetic scenarios (Section 7.1).

Scenario	n	Topo	Topo-QoS	GAT-N	GAT-N-QoS	HGT	HGT-QoS
AV System	16	0.297 $\pm$ 0.145	0.723 $\pm$ 0.099	0.761 $\pm$ 0.065	0.574 $\pm$ 0.173	0.788 $\pm$ 0.048	0.679 $\pm$ 0.051
Enterprise	60	0.415 $\pm$ 0.046	0.805 $\pm$ 0.023	0.577 $\pm$ 0.207	0.860 $\pm$ 0.021	0.879 $\pm$ 0.020	0.890 $\pm$ 0.014
Financial Trading	12	0.268 $\pm$ 0.139	0.700 $\pm$ 0.069	0.821 $\pm$ 0.045	0.804 $\pm$ 0.088	0.861 $\pm$ 0.028	0.777 $\pm$ 0.103
Healthcare	10	-0.155 $\pm$ 0.094	0.768 $\pm$ 0.072	0.587 $\pm$ 0.173	0.580 $\pm$ 0.145	0.635 $\pm$ 0.103	0.645 $\pm$ 0.145
Hub-and-Spoke	14	0.285 $\pm$ 0.099	0.401 $\pm$ 0.124	0.406 $\pm$ 0.062	0.043 $\pm$ 0.230	0.527 $\pm$ 0.011	0.296 $\pm$ 0.212
IoT Smart City	40	-0.058 $\pm$ 0.058	0.071 $\pm$ 0.070	0.861 $\pm$ 0.017	0.676 $\pm$ 0.125	0.885 $\pm$ 0.012	0.847 $\pm$ 0.030
Microservices	18	0.352 $\pm$ 0.134	0.698 $\pm$ 0.082	0.540 $\pm$ 0.094	0.452 $\pm$ 0.100	0.476 $\pm$ 0.126	0.469 $\pm$ 0.137
Mean	—	0.201	0.595	0.650	0.570	0.722	0.658

Table 7: Paired Wilcoxon signed-rank tests across scenarios ($n = 7$, two-sided). At $n = 7$ the smallest attainable two-sided p is 0.0156, which is above $\alpha/5 = 0.010$; flagged results should be read as uncorrected and exploratory (Section 6.3).

Comparison	$\Delta\rho$	Won	Wilcoxon W	p -value	Significance
HGT-QoS vs. Topo	+0.457	7/7	0.0	0.0156	Statistically Significant ($p < 0.05$)
Topo-QoS vs. Topo	+0.394	7/7	0.0	0.0156	Statistically Significant ($p < 0.05$)
HGT-QoS vs. GAT-N-QoS	+0.088	6/7	2.0	0.0469	Statistically Significant ($p < 0.05$)
HGT vs. GAT-N	+0.072	6/7	5.0	0.156	Not significant
HGT vs. HGT-QoS	+0.064	5/7	5.0	0.156	Not significant
HGT-QoS vs. Topo-QoS	+0.063	3/7	12.0	0.813	Not significant
GAT-N-QoS vs. Topo-QoS	-0.025	3/7	10.0	0.578	Not significant

7.2.1. Out-of-Distribution (LOSO) Generalization

In inductive Leave-One-Scenario-Out (LOSO) cross-validation, models are evaluated on their capacity to predict cascading criticality over completely unseen system topologies:

Table 8: Inductive Leave-One-Scenario-Out (LOSO) evaluation, Application population, twelve folds. Rows are grouped by role: training-free structural baselines, learned predictors on a shared native substrate, and the RM/ $Q(v)$ diagnostic reference of Section 1.2. All four learned variants receive the identical training set of $N - 1$ scenarios, the identical depth, and the identical model-selection rule (Section 6.3); they differ only in typing and edge channel. Each fold score is the mean over the five seeds; **Fold** σ is the population standard deviation of those twelve fold means, **Seed** σ is the median within-fold standard deviation across seeds (zero by construction for the deterministic baselines), and the 95% interval is a non-parametric bootstrap over folds ($B = 2,000$). Measured before the corpus repair of Section 7.1.

Predictor / Reference	Mean LOSO ρ	95% CI	Fold σ	Seed σ	Critical-Set $F_1@K$	Requires Training
<i>Training-free structural baselines</i>						
Topo	0.348	[0.272, 0.419]	0.132	—	0.372	No
Topo-QoS	0.601	[0.507, 0.680]	0.152	—	0.375	No
<i>Learned predictors (shared native substrate, matched training set)</i>						
GAT-N	0.600	[0.528, 0.667]	0.122	0.037	0.421	Yes
GAT-N-QoS	0.577	[0.510, 0.642]	0.115	0.012	0.440	Yes
HGT	0.664	[0.592, 0.732]	0.127	0.055	0.428	Yes
HGT-QoS	0.659	[0.585, 0.727]	0.129	0.025	0.439	Yes
<i>Diagnostic reference — not a ranking model</i>						
RM / $Q(v)$	0.225	[0.148, 0.290]	0.124	—	0.318	No

Twelve LOSO folds are reported, comprising the eleven synthetic evaluation scenarios and the ATM case study from Section 7.4. In each fold, one scenario is held out for zero-shot testing while the model is trained exclusively on the remaining eleven. All variants are evaluated on the identical Application node set per fold (Section 6.3); paired Wilcoxon tests are conducted across the twelve folds, where the smallest attainable two-sided p is 0.00049. Per-fold evaluated populations range from 26 to 300 Application nodes, so $K = \text{round}(0.20 |V_{\text{app}}|)$ ranges from 5 to 60. On $F_1@K$, HGT-QoS beats Topo-QoS in 7 of 12 folds ($\Delta = +0.064$, $W = 23.0$, $p = 0.233$) and GAT-N-QoS in 6 of 12 ($\Delta = -0.001$, $W = 17.0$, $p = 0.322$): critical-set identification separates the learned models from neither the training-free baseline nor untyped learning.

Label-noise ceiling. These correlations are bounded by the reproducibility of the target they are scored against. Re-running the ground-truth oracle across the five seeds gives a test-retest rank correlation of 0.97–1.00 in six of the seven original synthetic scenarios; the exception is Microservices at $\rho = 0.807$, which sets the effective ceiling for cross-fold means. HGT’s $\rho = 0.664$ therefore recovers roughly four fifths of the attainable signal, and no predictor in Table 8 can exceed the reproducibility of its own labels. Top- K critical sets are the noisier construct: their cross-seed Jaccard falls to 0.44 (Microservices) and 0.71 (AV, Financial Trading), which is part of why the $F_1@K$ margins above are less stable than the ranking margins.

Key Insights for RQ1:

- 1. Typed learning is the best configuration, but its margin is over untyped learning, not over the QoS baseline.** HGT leads all predictors on out-of-distribution ranking ($\rho = 0.664$), and the typed-vs-untyped margin excludes zero decisively (Section 7.3). Against the training-free *Topo-QoS* baseline, however, the margin is +0.063 with a 95% bootstrap interval of $[-0.012, +0.136]$ that contains zero, won in 9 of 12 folds ($W = 21.0$, $p = 0.176$).
- 2. A QoS-weighted structural score is a genuinely strong baseline.** Topo-QoS reaches $\rho = 0.601$ zero-shot, beating unweighted Topo on all twelve folds (+0.253, $W = 0.0$, $p = 0.0005$) and statistically indistinguishable from untyped graph learning (GAT-N -0.002 , $W = 30.0$, $p = 0.519$; GAT-N-QoS -0.024 , $W = 35.0$, $p = 0.791$). Any claim that graph learning is required must be made against this baseline rather than against unweighted centrality.
- 3. Power is not the limiting factor.** At $n = 12$ the signed-rank design tolerates four lost folds and still reaches $\alpha = 0.05$, provided the losses are the smallest in magnitude. HGT-QoS’s three losses are not: Enterprise (-0.346) and Microservices (-0.173) are the second- and fourth-largest $|\Delta|$ in the fold set, which is what holds W at 22.0. Enlarging the corpus further will not resolve this; the two inverted folds must be understood instead (Section 7.3.2).

4. **The explanation layer is weakly predictive, not noise.** $RM/Q(v)$ achieves $\rho = 0.225$ under distribution shift — outperforming unweighted topological baselines is not claimed, since Topo reaches 0.348, but it remains well above chance while providing interpretable architectural diagnostics without training (Section 5).

7.2.2. Nested Model Selection

The Table 8 figures fix one configuration for every fold. Because the representation choices in Section 4.1 (within-graph rank normalisation of features and of labels) and the network depth are free parameters, we additionally report a nested cross-validation in which they are selected *inside* each fold’s training set: for outer fold k , each of eight configurations is scored on two inner folds drawn from the $N - 1$ training scenarios, and the winner is trained on all $N - 1$ and evaluated once on k . The held-out scenario takes no part in selection — not in early stopping, not in checkpoint choice, not in configuration choice — and the harness asserts this per fold.

These nested-selection figures, like Table 8, predate the corpus repair of Section 7.1. Under nested selection HGT-QoS reaches $\rho = 0.716 \pm 0.117$, a margin of $+0.115$ over Topo-QoS won in 10 of 12 folds ($W = 14.0$, $p = 0.0522$; Holm-adjusted 0.104 across the two pre-registered comparisons). This is the strongest ranking result in the study and it still does not reach $\alpha = 0.05$. We report it as such: the margin is practically meaningful and the evidence is short of conventional significance on a twelve-domain corpus.

Two observations follow. First, rank normalisation earns its place: label rank normalisation is selected in 9 of 12 folds and feature rank normalisation in 8 of 12, which is what one expects when the reported metric is itself a rank correlation. Second, selection is unstable — six distinct configurations win across twelve folds, with depth 3 chosen in only 7 — so no single configuration is recommended on this evidence, and the nested figure should be read as the performance of the *procedure*, not of a shipped setting.

Figure 3 provides an overview of these results, summarizing out-of-distribution rank correlation, critical-set identification, and cross-oracle agreement.

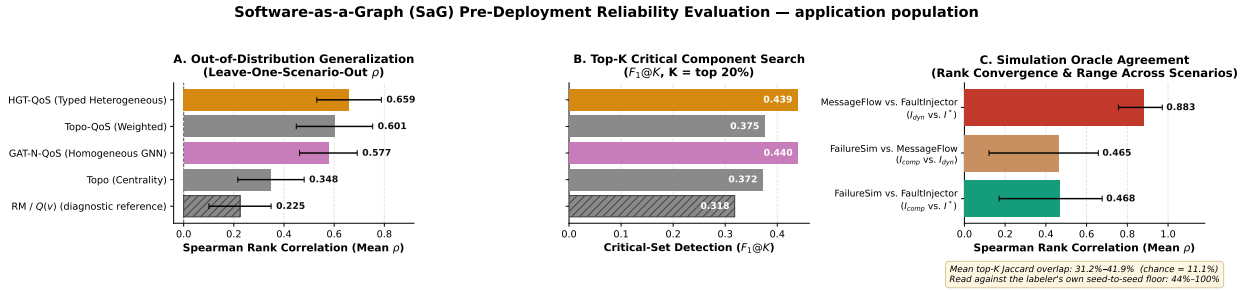


Figure 3: Results at a glance, evaluated on the Application population. **(A)** Out-of-distribution rank correlation per variant across twelve LOSO folds (whiskers denote σ). **(B)** Critical-set identification at $K = 20\%$. **(C)** Pairwise agreement across the three simulation oracles.

7.3. RQ2: Value of Typed Heterogeneity

To evaluate the specific contribution of node and edge typing, we contrast the relation-specific Heterogeneous Graph Transformer against the homogeneous Graph Attention Network on the shared native multigraph substrate, with the identical training set, depth and model-selection rule:

- **In-Distribution Fitting (Table 6):** Typed message passing carries a modest advantage on familiar architectures ($\rho = 0.658$ for HGT-QoS vs. 0.570 for GAT-N-QoS; $\Delta\rho = +0.088$, $W = 2.0$, $p = 0.0469$, won in 6/7 scenarios). At $n = 7$ this sits one fold away from non-significance and should be read as suggestive.
- **Out-of-Distribution Generalization (LOSO, Table 8):** Under inductive distribution shift, the typed advantage is unambiguous. HGT-QoS outperforms GAT-N-QoS by $+0.082$ ($\rho = 0.659$ vs. 0.577),

winning *all twelve* folds ($W = 0.0$, $p = 0.0005$; 95% CI $[+0.064, +0.102]$). The unweighted pair replicates it exactly: HGT over GAT-N by $+0.064$, 12/12 folds, $W = 0.0$, $p = 0.0005$, CI $[+0.045, +0.082]$.

A small effect, measured cleanly. Typing is the only comparison in this study that wins every fold at the attainable p -floor, under both edge channels, in both evaluation regimes. Its effect size is nonetheless an order of magnitude smaller than the corpus-level differences that motivate the framework: $+0.08$ against Topo-QoS’s own $+0.25$ over unweighted centrality. Relation-specific parameters (distinguishing PUBLISHES_TO message dissemination from RUNS_ON host placement) do capture invariant semantics of failure propagation that transfer across unseen systems — consistently, reproducibly, and by a small amount.

7.3.1. Harness Attribution of the Typed–Untyped Margin

An earlier version of this evaluation reported the LOSO typed–untyped margin as $+0.246$ rather than $+0.082$. That figure was an artifact of three defects in the evaluation harness, each of which advantaged the typed model, and we report the attribution rather than the corrected number alone.

The homogeneous branch trained on the single largest training scenario while the typed branch received all $N - 1$ scenarios through its inductive-graph channel; the network depth was derived from the size of the largest *training* graph, so it varied with which scenario was held out; and early stopping selected checkpoints on a within-scenario validation split of a training graph, which selects for in-distribution fit under a protocol whose purpose is distribution shift. Correcting these in sequence, on the identical twelve-fold corpus and cache, moves the untyped GAT-N baseline as shown in Table 9.

Table 9: Attribution of the untyped baseline’s movement to individual harness corrections (GAT-N, twelve LOSO folds, five seeds, identical cache throughout). Arm A runs the pre-correction code; arms B and C add corrections cumulatively. Measured before the corpus repair of Section 7.1.

Arm	Configuration	Mean LOSO ρ	Δ vs. previous
A	Pre-correction harness	0.115 ± 0.118	—
B	+ matched training set, constant depth	0.414 ± 0.133	$+0.299$
C	+ held-out-scenario model selection	0.600 ± 0.122	$+0.185$

Neither correction has anything to do with typing, and together they account for $+0.484$ of the $+0.514$ total movement; the residual is attributable to the corpus growing from eight folds to twelve. The typed advantage that survives is the $+0.082$ of Section 7.3. We record this because the replication package ships with the paper: a reader who reruns the pre-correction harness will reproduce 0.115, and the discrepancy would otherwise be unexplained.

7.3.2. Where Typed Learning Fails, and How It Can Be Detected

HGT-QoS loses to Topo-QoS on exactly two folds, and they are the two that hold the RQ1 comparison below significance: Enterprise ($\rho = 0.492$ vs. 0.839) and Microservices (0.452 vs. 0.626). Both are folds on which the training-free baseline is unusually strong, which suggests the model is discarding structural signal the baseline retains rather than failing to learn.

The two folds share a measurable signature. Let $\hat{\sigma}$ denote the standard deviation of the model’s own predicted scores over the held-out Application population — a quantity computable at inference time, without labels. Enterprise and Microservices carry the two lowest values in the corpus ($\hat{\sigma} = 0.074$ and 0.128 , against a corpus median of 0.155 and a truth spread of 0.297 and 0.476 respectively): on precisely the graphs where it loses, the model’s output collapses toward a constant. Across the twelve folds, $\hat{\sigma}$ correlates with the margin over Topo-QoS at $\rho_s = +0.54$ ($p = 0.071$) for HGT-QoS and $+0.52$ ($p = 0.085$) for HGT, weakening to $+0.38$ ($p = 0.226$) for untyped GAT-N. At $n = 12$ these are suggestive rather than established, and we present them as a diagnostic hypothesis rather than a result.

Graph size does not explain the failure: the correlation between fold size and margin is -0.03 , and the two largest wins are on the third- and fifth-largest graphs (IoT Smart City, $+0.463$; Industrial SCADA, $+0.172$). Nor is the failure merely one of compression — on Enterprise the model inverts the extremes, ranking four maximum-impact Applications in the bottom third and four zero-impact Applications in the top fifth.

Feature *scale* is a different matter from graph size, and part of the Enterprise deficit is recoverable. Under the nested selection of Section 7.2.2, Enterprise gains more than any other fold ($\rho = 0.492 \rightarrow 0.668$, $+0.176$, against a corpus mean gain of $+0.057$), and its selected configuration includes within-graph feature rank normalisation — the transform motivated by the cross-scenario scale drift of `results/feature_shift_diagnostic.md`, and the one whose effect should be largest on the corpus’s biggest graph. We stop short of attributing the recovery to that axis: the folds selecting feature rank normalisation gain $+0.061$ on average against $+0.047$ for those that do not, a separation far too small to isolate one axis from depth and label normalisation in an eight-configuration search. What the nested result does establish is that roughly half the Enterprise gap ($-0.347 \rightarrow -0.171$ against Topo-QoS) is a configuration artifact rather than a property of typed learning, while the other half, and the Microservices deficit ($+0.046$ recovered of -0.174), survive selection and remain unexplained.

The practical consequence is that this failure mode announces itself. Because $\hat{\sigma}$ requires no ground truth, a deployment can determine that a trained ranker is untrustworthy on a given architecture before any prediction from it is acted upon — which is a more useful property for a pre-deployment gate than a marginally higher mean correlation would have been.

7.3.3. Empirical Edge-Removal Analysis

We further evaluated edge criticality by simulating the removal of individual edges while keeping both endpoint components operational. The 50 highest-ranked candidate edges in the `av_system` topology comprised 35 `RUNS_ON`, 11 `CONNECTS_TO`, 3 `SUBSCRIBES_TO` and 1 `PUBLISHES_TO` edge. Exactly 4 removals produced non-zero downstream cascade impact, and they were exactly the 4 communication channels: every host-placement and inter-host network edge in the pool was inert, while every publish–subscribe edge in it was not. The effect is nonetheless small in magnitude — the largest single-edge impact was 0.00504 — so the finding is that severing a message channel degrades reachability slightly whereas relocating a component or dropping a host link does not, not that any one channel is load-bearing. With only 4 communication edges in the candidate pool this is a descriptive observation on one topology rather than a tested claim, and we report it as such.

7.4. RQ3: Ablations and Sensitivity Analysis

Role of ρ in Sensitivity Sweeps. The parameter sweeps below utilize RM’s rank correlation against $I^*(v)$ strictly as a *sensitivity probe* to quantify parameter leverage relative to between-model margins.

7.4.1. QoS Feature Ablation

To isolate the specific empirical contribution of the 7-dimensional continuous-categorical QoS edge features (Section 4.1.1), we evaluated **HGT**, an un-augmented ablation of HGT-QoS whose edge features contain only scalar coupling and relation one-hot encodings.

Under the corrected harness of Section 7.3.1, ablating the QoS edge encodings has *no detectable effect on ranking accuracy*. Mean LOSO rank correlation is $\rho = 0.659$ with the encodings and 0.664 without them ($\Delta\rho = -0.005$, won in 6 of 12 folds, $W = 33.0$, $p = 0.677$; 95% bootstrap CI $[-0.043, +0.035]$). On the homogeneous architecture the direction is the same and slightly larger: GAT-N-QoS 0.577 against GAT-N 0.600 ($\Delta\rho = -0.023$, 3 of 12 folds, $W = 19.0$, $p = 0.129$). In-distribution the pattern repeats (HGT 0.722 vs. HGT-QoS 0.658, 5/7, $p = 0.156$). We previously reported this ablation as a $+0.169$ advantage for the QoS-augmented model; that figure was produced under the defective harness documented in Section 7.3.1 and does not survive its correction.

What the QoS encoding does still buy is training stability rather than accuracy: the median within-fold standard deviation across the five seeds is 0.025 for HGT-QoS against 0.055 for HGT, and 0.012 for GAT-N-QoS against 0.037 for GAT-N — the QoS-augmented variant is roughly twice as reproducible across initialisations in both architectures.

This ablation is superseded and must be re-run. The measurement above was taken on the corpus as it stood before the QoS generation defect of Section 7.1 was repaired, and the explanation we previously gave for it no longer holds. On that earlier corpus, seven of the twelve scenarios declared an identical (**RELIABLE**,

TRANSIENT_LOCAL) pair on every topic, so reliability, durability, priority and the heterogeneity flag were constant across every QoS-bearing edge of those graphs and could not discriminate between their components by construction. The near-zero $\Delta\rho$ was therefore measured where its treatment was largely inert, and we read it as uninformative rather than as evidence against the encoding.

That degeneracy is gone. The generator now samples each topic’s profile from the per-scenario `qos_stats` distributions rather than from a name-pattern lookup, and the diagnostic re-run on the shipped corpus (`reproduce/qos_corpus_diagnostic.py`; artifact `results/qos_corpus_diagnostic.json`) reports **zero degenerate scenarios of twelve**, with 13 to 30 distinct QoS triples per scenario and a modal share between 9% and 30%. The condition that made the ablation unmeasurable has been removed, but the ablation itself has not yet been repeated: $\Delta\rho = -0.005$ is a pre-repair number, and we state it as such rather than restating it as a finding about the corpus that ships with this paper. Re-running it is the first item of the regeneration described in Section 7.1, and we make no prediction here about which way it will move.

One half of the earlier diagnostic does survive the repair and is unchanged: three of the seven encoded dimensions (`has_deadline`, `deadline_ns_log`, `max_blocking_ms_log`) remain identically zero in all twelve scenarios, because QoSPolicy carries no deadline or blocking fields for the generator to emit. Those three slots have never carried a bit of information in any model reported here, and populating them requires extending the QoS model itself (Section 8.3).

7.4.2. Topic-Weight Coefficients (β , α , ψ)

We swept (β, α, ψ) over a grid spanning the entire simplex to evaluate sensitivity to topic coupling weights, with results detailed in Table 10:

Table 10: Sensitivity of topic-weight ordering and downstream rank correlation against $I^*(v)$ to coefficients (β, α, ψ) (Application population, mean over 7 scenarios, 375 topics). Measured before the corpus repair of Section 7.1.

(β, α, ψ)	ρ of $w(t)$ vs. shipped	Topo-QoS ρ	RM ρ
$(0.75, 0.15, 0.10)$ <i>shipped</i>	1.000	0.604	0.267
$(0.90, 0.05, 0.05)$	0.966	0.600	0.273
$(0.85, 0.15, 0.00)$	0.950	0.608	0.268
$(1.00, 0.00, 0.00)$	0.852	0.618	0.268
$(0.60, 0.20, 0.20)$	0.970	0.604	0.261
$(0.50, 0.25, 0.25)$	0.942	0.603	0.259
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ <i>uniform</i>	0.870	0.608	0.256
Spread over grid	—	0.018	0.017

Topic-weight coefficients are not load-bearing: the ordering of $w(t)$ never falls below $\rho = 0.852$ against the default, and downstream rank correlations vary by at most 0.018 (Topo-QoS) and 0.017 (RM).

7.4.3. Intra-Dimension AHP Weight Shrinkage

We evaluated the sensitivity of the RM attribution baseline across shrinkage parameter $\lambda \in [0, 1]$, blending internal term weights from a uniform prior ($\lambda = 0$) to raw AHP judgment ($\lambda = 1$), as summarized in Table 11:

Table 11: Sensitivity of RM rank correlation against $I^*(v)$ to AHP shrinkage parameter λ (Application population, mean over 7 scenarios). Measured before the corpus repair of Section 7.1.

λ Setting	0.00 (Uniform)	0.50	0.70 (Default)	0.80	1.00 (Raw AHP)
Mean Rank Correlation (ρ)	0.348	0.291	0.267	0.256	0.232

Rank correlation decreases monotonically as weights transition from uniform toward raw AHP ($\rho = 0.348 \rightarrow 0.232$, Figure 4). Elicited AHP weights provide transparent, auditable domain attribution rather than optimizing rank correlation. We retain $\lambda = 0.70$ on that basis.

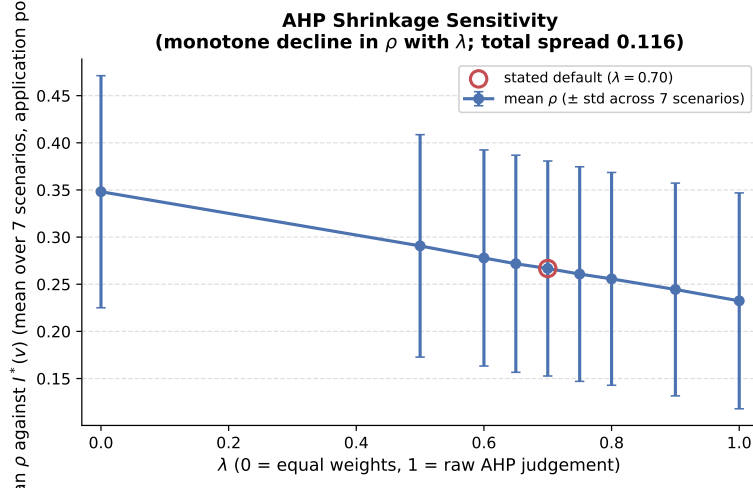


Figure 4: Sensitivity of RM composite rank correlation against $I^*(v)$ across AHP shrinkage λ (Application population, mean over 7 scenarios). Measured before the corpus repair of Section 7.1.

7.4.4. Joint Sensitivity Across All Ten Weight Constants

To assess interactions, we swept all ten weight constants jointly across six scenarios using Morris elementary-effects screening [81, 82] (a computationally efficient alternative to variance-based global sensitivity analysis [83, 84]), with factor influences reported in Table 12:

Table 12: Morris elementary-effects screening [81, 82] across ten weight constants, ranked by influence (μ^*) on mean ρ (6 scenarios, 10 trajectories, 110 evaluations). Measured before the corpus repair of Section 7.1.

Factor	μ^*	σ
λ (AHP shrinkage)	0.124	0.077
r_α	0.096	0.035
α	0.019	0.034
w_{rel}	0.019	0.018
β	0.017	0.023
w_{dur}	0.014	0.017
w_{prio}	0.013	0.018
ψ	0.012	0.015
p (power-mean exponent)	0.005	0.005
γ (fan-out)	0.001	0.002

Morris screening confirms that λ and r_α are the primary drivers, while topic and QoS sub-weights reside in a modest band ($\mu^* \in [0.012, 0.019]$). Dirichlet simplex sampling over 100 draws confirms tight stability: mean $\rho = 0.243$ (standard deviation 0.006, 90% interval [0.231, 0.253], mean top-20% Jaccard 0.825).

7.4.5. Convergent Validity Over Simulation Oracles

We evaluated inter-oracle agreement across $I^*(v)$ (**FaultInjector**), $I_{\text{comp}}(v)$ (**FailureSimulator**), and $I_{\text{dyn}}(v)$ (**MessageFlowSimulator**) over seven scenarios, summarized in Table 13: Ordering converges strongly between the queue-flow simulator and topological cascade oracle ($\rho = 0.907$). Top- K set agreement is more conservative (Jaccard 0.486), bounded by discrete threshold sensitivity in non-linear cascades.

Table 13: Inter-oracle agreement across simulation paradigms (chance top- K Jaccard is 0.111), measured on the regenerated corpus. Two independent corrections separate these figures from those an earlier draft reported, and they act in opposite directions. First, I_{comp} was computed without establishing the discrete-event baseline its flow-disruption criterion is measured against, so it silently carried three of its four terms; priming it moved I_{comp} *further* from both other oracles (ρ 0.468 $\rightarrow$ 0.438 against I^* , 0.465 $\rightarrow$ 0.439 against I_{dyn}) and left the I_{dyn}/I^* row identical to four decimal places, as it must, since neither of those oracles reads the composite. Second, the corpus regeneration that gave topics genuinely varied QoS moved every row: the behavioural/cascade agreement rose (0.883 $\rightarrow$ 0.907) while both I_{comp} pairs fell further (0.438 $\rightarrow$ 0.425 and 0.439 $\rightarrow$ 0.427). The negative lower bounds on the I_{comp} rows are a single scenario, **hub_and_spoke**, and Section 7.4.6 shows that sign is an artifact of tied zeros rather than a disagreement about ordering.

Oracle pair	Mean ρ (range)	Mean τ	Jaccard@ K	Tie-robust
I_{dyn} vs. I^*	0.907 (0.748–0.985)	0.788	0.486	0.509
I_{comp} vs. I^*	0.425 (−0.044–0.654)	0.311	0.240	0.258
I_{comp} vs. I_{dyn}	0.427 (−0.037–0.654)	0.312	0.276	0.275

7.4.6. Zero-Inflation Sensitivity of the Agreement Figures

Spearman ρ over a population where many components are tied at exactly zero is driven substantially by how those ties are handled, and $I^*(v)$ is heavily zero-inflated: 19 to 106 Applications per scenario carry exactly zero cascade impact, whereas $I_{\text{comp}}(v)$ has no exact zeros at all. We therefore report, alongside each full-population figure, $\rho_{>0}$ — the same correlation restricted to components both oracles score strictly positive. It is a sensitivity bound, not a replacement: for a component the simulator actually injected, zero impact is a real measurement ("its failure reaches nobody"), and dropping it would be a results-favourable filter.

The bound changes the reading of two scenarios, in opposite directions. On **hub_and_spoke**, I_{comp} and I^* correlate at $\rho = -0.044$ over the full population but $\rho_{>0} = +0.263$ over the 22 components both score positive: the apparent *sign* disagreement is an artifact of tied zeros, not evidence that the oracles order active components inversely. On **microservices**, the movement runs the other way — $\rho = 0.461$ falls to $\rho_{>0} = 0.257$, so a substantial part of that scenario’s agreement is the two oracles concurring on which components are inert rather than on how the active ones rank. Averaged over the seven scenarios the two summaries are close ($\rho = 0.425$, $\rho_{>0} = 0.447$), which is precisely why the per-scenario figures matter: the mean conceals compensating movements of 0.2–0.3 in both directions.

7.4.7. Domain-Specific Weighting and Threshold Sensitivity

Sweeping composite reliability weight $w_R \in [0, 1]$ moves mean ρ by only 0.024 (0.341 $\rightarrow$ 0.365), with mean Kendall correlation $\tau = 0.974$ between domain and static rankings. Within that narrow band the domain-derived weighting does not improve on the alternatives it replaces — mean ρ is 0.347 (domain-derived), 0.349 (equal), and 0.353 (static) — so domain reweighting should be understood as an attributional device that expresses criticality in stakeholder terms, not as a ranking-accuracy mechanism. No weighting confined to this parameter could move ρ appreciably. Sweeping cascade propagation thresholds revealed higher sensitivity ($\Delta\rho = 0.102$), with ranking performance plateauing above 0.35.

7.4.8. Anti-Pattern Detection and Node-Type Stratification

Validated against $I_{\text{comp}}(v)$, the rule-based anti-pattern catalog reaches mean precision 0.239 and recall 0.900 ($F_1 = 0.378$), but the recall must be read against a flag rate of 94.2%: the 18 active detectors implicate almost everything, so recall near 0.9 is close to what indiscriminate flagging yields and precision 0.239 is roughly the base rate. Chance-corrected agreement confirms this and worsens with scale — over the ATM sweep (29 to 444 components, five seeds) Cohen’s κ falls monotonically from 0.118 to -0.045 , i.e. to no better than chance on the largest graph. We report the catalog as a coarse, deliberately over-inclusive triage filter, *not* a validated detector; every quantitative critical-set claim in this paper rests on the ranking models of Sections 7.2–7.3 instead. A nineteenth detector, **DEEP_PIPELINE**, is excluded outright: enumerating every simple source-to-sink path, it emitted 247,761 findings without terminating on a 29-component fixture.

Stratification vs. Pooling: Measured against the composite oracle $I_{\text{comp}}(v)$ over the eight scenarios of the detection benchmark, stratified RM rank correlations are $\rho = 0.503$ (Application, 8 scenarios), 0.395 (Broker,

6 scenarios), and 0.142 (Node, 8 scenarios), while pooled correlation across all types is $\rho = 0.028$. These figures are not comparable to the RM values elsewhere in this paper and should not be read as the same quantity: Table 11’s $\rho = 0.267$ scores RM in-sample against $I^*(v)$ over the seven synthetic scenarios, and Table 8’s $\rho = 0.225$ scores it zero-shot against $I^*(v)$ over the twelve LOSO folds. Oracle, corpus and protocol all differ; the contrast drawn here is strictly between the stratified and pooled readings of one benchmark. Pooling node types triggers Simpson’s paradox by mixing distinct base rates. Classifying components within node types changes the criticality tier of 62.8% of components (with 19.0% crossing the critical boundary), confirming that evaluation and gating must operate strictly per-type. On that same pooled benchmark, unweighted degree centrality attains $\rho = 0.166$ and $F_1 = 0.413$ against RM’s 0.028 and 0.349 — a further reason to read $Q(v)$ as an attribution instrument rather than as a ranking model.

7.4.9. HGT Attention Weight Analysis

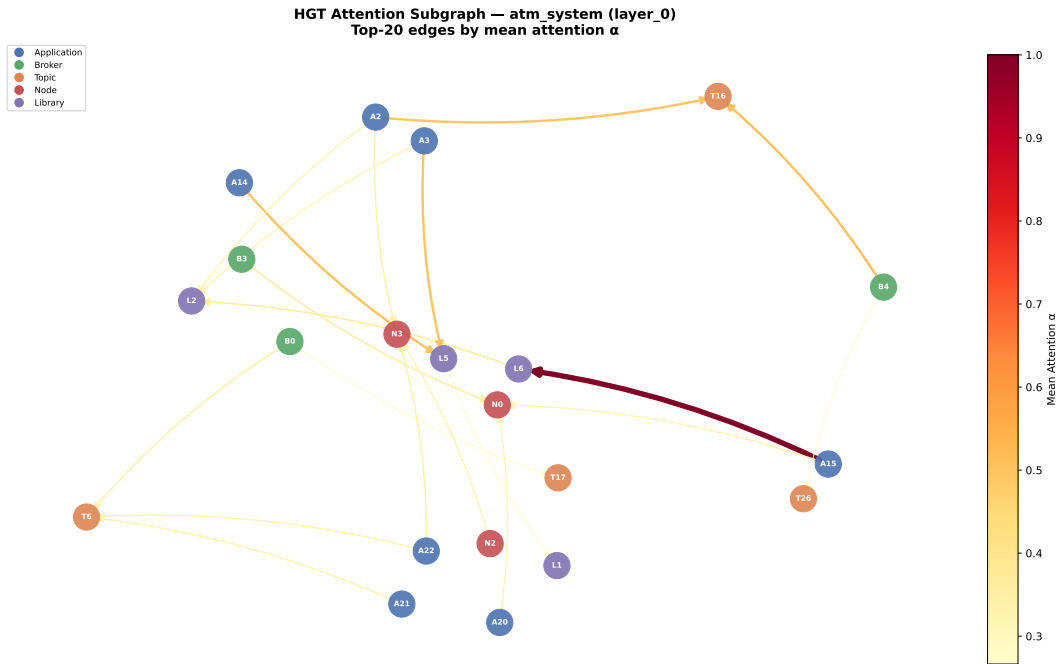


Figure 5: Relational attention extracted from the trained HGT on the ATM case study. Peak attention focuses on **USES** (Application $\rightarrow$ Library) and **ROUTES** (Broker $\rightarrow$ Topic) channels.

Aggregated by relation type over the ATM case study, first-layer mean attention orders **USES** into libraries (0.227, 0.215) above **ROUTES** (0.194), **SUBSCRIBES_TO** (0.176), **PUBLISHES_TO** (0.163) and **RUNS_ON** (0.153), with the same ordering to within 0.02 in layers two and three. Two cautions bound what this supports. The spread across all eight relation types is narrow (0.15–0.23), so this is a tendency, not a separation; and because α is a softmax over each destination’s incoming edges, mean α is driven substantially by in-degree — the top-ranked **Library** $\rightarrow$ **Library** relation carries 4 edges, and the largest weight in the graph ($\alpha_{uv} = 1.00$) sits at a destination of in-degree one, where $\alpha = 1$ holds by construction. Figure 5 is therefore a qualitative illustration on one topology at one seed without a significance test, not evidence for why typing helps.

7.5. RQ4: Real-World Distributed Architecture Validation

We evaluated the framework on five open-source distributed systems transcribed from their public repositories by dedicated architectural adapters, with results presented in Table 14. We note the scope of this evaluation before reporting it. Online Boutique is a vendor demonstration application and Train-Ticket

an academic benchmark, Home Assistant is an authentic open-source IoT automation platform, and EdgeX Foundry is an industrial edge computing reference implementation; each is an abstraction of its upstream repository rather than a complete transcription; and their labels come from the same simulation oracle used throughout, so what follows tests topological generalization, not agreement with observed field failures. Two further limits bound what this table can support. First, the predictor scored in Table 14 is the deterministic explanation layer $Q(v)$ together with the two training-free topological baselines, and it is scored against $I_{\text{comp}}(v)$; the learned model is evaluated separately in Section 7.5.1 against $I^*(v)$, and the two are not commensurable. Second, the labels are strongly tied at zero in several architectures — 13 of 32 Applications in Autoware, 14 of 22 in Cloud Microservices, 27 of 41 in Train-Ticket, and 12 of 22 in EdgeX carry zero simulated impact (the complement of the “Impactful Apps” column), whereas Home Assistant exhibits much lower zero-inflation with 17 of 24 applications actively participating in failure cascades. Spearman ρ over populations that are heavily tied is dominated by how those ties are handled; the values below use midrank ties throughout.

Table 14: Zero-shot evaluation on open-source reference systems. The scored predictor is the deterministic explanation layer RM / $Q(v)$ (Section 5), not a trained network: these runs invoke no GNN checkpoint. $\pm$ therefore denotes variance of the stochastic cascade oracle over the five simulation seeds {42, 123, 456, 789, 2024}, not training variance; $Q(v)$ itself is deterministic. “Gain vs. Deg” is $|\rho_Q| - |\rho_{\text{deg}}|$. Ground truth is $I_{\text{comp}}(v)$ (FailureSimulator); the learned-model evaluation of Table 15 uses $I^*(v)$ (FaultInjector) and the two columns must not be read against each other. These are simulator-derived labels, not observed production incidents.

Real-World Architecture	$ V $	$ V_{\text{app}} $	Spearman ρ	Kendall τ	App $F_1@K$	Pooled $F_1@K$	Impactful Apps	Gain vs. Deg
Autoware.universe (ROS 2)	75	32	0.685 $\pm$ 0.010	0.513	0.333	0.800	19 / 32	+0.357
Cloud Microservices Mesh	60	22	0.778 $\pm$ 0.001	0.639	0.500	1.000	8 / 22	+0.014
Train-Ticket Booking Mesh	90	41	0.759 $\pm$ 0.001	0.605	0.625	1.000	14 / 41	+0.264
Home Assistant (Smart Home)	63	24	0.514 $\pm$ 0.016	0.373	0.250	0.667	17 / 24	+0.289
EdgeX Foundry (Industrial IoT)	63	22	0.800 $\pm$ 0.015	0.563	0.000	1.000	10 / 22	+0.427

Key Insights for RQ4:

- The explanation layer ranks unseen architectures well, but this is not a learned-transfer result.** RM / $Q(v)$ attains $\rho = 0.800$ on EdgeX Foundry, 0.778 on Cloud Microservices, 0.759 on Train-Ticket, 0.685 on Autoware.universe, and 0.514 on Home Assistant. Because $Q(v)$ is a closed-form scoring function rather than a fitted model, applying it to a new architecture involves no transfer in the inductive sense of Section 7.3: nothing was trained, and no distribution shift is being survived. What this table establishes is that the deterministic attribution of Section 5 remains informative on architectures we did not generate. The learned model is a separate question, answered in Section 7.5.1. Table 14 is not commensurable with Table 8’s RM column ($\rho = 0.225$), and the difference is primarily one of oracle rather than of architecture: Table 14 scores $Q(v)$ against $I_{\text{comp}}(v)$ (FailureSimulator), whereas Table 8 scores it against $I^*(v)$ (FaultInjector). Scored against $I^*(v)$ on these same five systems, RM attains $\rho = 0.516$ (Table 15) — so roughly half of the apparent gap is the oracle change and the remainder is the real-versus-synthetic difference, with the tied-label structure described above contributing to both.
- Stratified vs. Pooled Critical Set Identification:** On the stratified Application population (V_{app} with $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$), $Q(v)$ identifies the top-20% critical services with $F_1@K = 0.333$ in Autoware, 0.500 in Cloud Microservices, 0.625 in Train-Ticket, 0.250 in Home Assistant, and 0.000 in EdgeX Foundry (where failure impact is dispersed broadly across active device services). When evaluated across the pooled multigraph ($K = \text{round}(0.20 \cdot |V|)$), the pooled $F_1@K$ reaches 0.800, 1.000, 1.000, 0.667, and 1.000, respectively. This disparity highlights the base-rate phenomenon discussed in Section 6.3: the passive infrastructure entities (Topics, Nodes, Libraries) carry constant zero failure impact in *these runs*, which were produced before the injector gained Topic and host-Node modes (Section 8.3). The pooled figures in this row are therefore a property of the run that produced them, not of the engine as it now stands; re-running Table 14 against the extended injector would change the pooled column and leave the stratified V_{app} column — the one we read — untouched. While the pooled metric confirms that $Q(v)$ cleanly separates critical active services from inert infrastructure, reporting stratified V_{app} metrics provides the more rigorous, non-inflated evaluation.

3. **Framework Acceptance Gates and Cross-Domain Disparities:** The framework ships a five-condition release gate ($\rho \geq 0.75$, $F_1 \geq 0.65$, SPOF $F_1 \geq 0.60$, fault-tolerance ratio ≤ 0.30 , prediction gain ≥ 0.02). EdgeX Foundry passes all five gates across all five seeds (100% pass rate), driven by strong rank correlation ($\rho = 0.800$), pooled $F_1 = 1.00$, and robust articulation detection (SPOF $F_1 = 0.80$). In contrast, the pass rate is zero on the other four architectures: Autoware fails the ρ and SPOF conditions, Cloud Microservices fails SPOF and prediction gain, Train-Ticket fails SPOF, and Home Assistant misses the ρ gate ($\rho = 0.514$) despite scoring a perfect SPOF $F_1 = 1.00$. This indicates that while structural rank ordering transfers well across distributed paradigms, absolute gating thresholds require domain-specific calibration.
4. **Predictive Advantage over Structural Heuristics:** The “Gain vs. Deg” column reports $Q(v)$ ’s rank-correlation margin over an unweighted degree centrality baseline (+0.427 on EdgeX Foundry, +0.357 on Autoware, +0.289 on Home Assistant, +0.264 on Train-Ticket, +0.014 on Cloud Microservices), demonstrating that hierarchical attribution (Section 5) consistently resolves topological fragility that degree centrality obscures across diverse paradigms. As before, Wilcoxon signed-rank tests compare errors across distinct scales and do not reach significance ($p \geq 0.33$), but we report them transparently.
5. **The QoS-weighted heuristic does not replicate its synthetic-corpus advantage here.** Both training-free baselines are computable on these systems — every topic in all five adapters carries declared `durability`, `reliability` and `transport_priority` profiles, and every derived `DEPENDS_ON` edge consequently carries a non-unit QoS weight. Scored on the identical Application population and ground truth, `Topo` attains $\rho = 0.346/0.928/0.588$ and `Topo-QoS` $0.362/0.884/0.522$ on Autoware, Cloud Microservices and Train-Ticket. QoS weighting does not universally improve ranking on non-generated architectures, consistent with the low- $w(t)$ -variance caveat of Section 6.2. Projection-based baselines assign constant zero to Applications carrying no derived `DEPENDS_ON` edge, accentuating sensitivity to tied ground truth.

7.5.1. Zero-Shot Transfer of the Learned Model

The typing result of Section 7.3 is established entirely on scenarios from our own generator. To test whether it survives contact with architectures we did not author, we trained HGT-QoS on all twelve synthetic scenarios and scored it zero-shot on the same five open-source systems, against $I^*(v)$ — the oracle it was trained on — over the same five seeds. The two training-free references were scored against the identical labels, population and node set, so Table 15 is a like-for-like comparison. It does not go in the learned model’s favour.

Table 15: Zero-shot transfer to the five open-source reference systems, scored against $I^*(v)$ (`FaultInjector`) on the Application population. HGT-QoS is trained on all twelve synthetic scenarios; $\pm$ is the standard deviation over the five training seeds. RM and Topo are training-free and were scored against the identical labels, population and node set. `Topo-QoS` is absent because it requires QoS-weighted betweenness on the projection graph, which this cache does not carry. $\hat{\sigma}$ is the spread of the model’s own predictions (Section 7.3.2). These figures use a different oracle from Table 14 and are not comparable to it.

Real-World Architecture	$ V_{\text{app}} $	RM ρ	Topo ρ	HGT-QoS ρ	HGT-QoS $F_1 @ K$	$\hat{\sigma}$
Cloud Microservices Mesh	22	0.777	0.891	-0.233 ± 0.180	0.000	0.091
Train-Ticket Booking Mesh	41	0.713	0.528	-0.049 ± 0.183	0.125	0.122
Autoware.universe (ROS 2)	32	0.357	0.307	0.060 ± 0.107	0.267	0.109
EdgeX Foundry (Industrial IoT)	22	0.470	0.534	0.200 ± 0.373	0.350	0.122
Home Assistant (Smart Home)	24	0.265	0.297	0.648 ± 0.228	0.600	0.151
Mean	—	0.516	0.511	0.125	—	—

The learned model does not transfer to these systems; the training-free baselines do. HGT-QoS averages $\rho = 0.125$ against 0.516 for RM and 0.511 for Topo, and it is negative on two of the five (Cloud Microservices -0.233 , Train-Ticket -0.049). This is not a failure of the oracle or of the transcription: the training-free scores were computed against the same labels on the same nodes and hold up at roughly 0.51. It is the learned model specifically failing to generalize beyond its generator, and going bimodal in the process —

Home Assistant, where it posts its best score in the study (+0.648, against 0.265 and 0.297 for the references), is the same model on the same weights.

One caveat bounds how much the references establish. RM and Topo correlate at $\rho = 0.66$ –0.92 on these five graphs, with top- K overlaps of 4/4 to 7/8, so they are not independent points of reference; on non-generated architectures RM largely restates articulation and betweenness rather than adding to them. What they jointly establish is that a training-free structural score ranks these systems, and the trained network does not.

The failure is detectable without labels. The $\hat{\sigma}$ diagnostic of Section 7.3.2 was derived on synthetic folds, and this is the first test of it off-corpus. All five systems fall below the synthetic median of $\hat{\sigma} = 0.155$, so a gate calibrated on the corpus rejects all five — correctly, since the learned model should not have been trusted on any of them. Across the five, $\hat{\sigma}$ orders with ρ at $\rho_s = +0.700$, ranking four of five correctly with Train-Ticket the inversion. Two limits apply: $n = 5$ makes this directional only, and the 0.155 threshold was measured before the corpus regeneration of Section 7.1, so the comparison crosses corpora and is indicative rather than exact.

Section 7.2 showed the learned model failing to separate from a training-free baseline in distribution; here it loses to one outright. Section 8.1 draws the practical consequence.

7.6. RQ5: Analysis Cost, Computational Sustainability, and CI/CD Feasibility

RQ5 quantifies computational overhead and sustainability during CI/CD evaluation, with per-stage latencies reported in Table 16:

Table 16: Per-stage latency of the inference pipeline across scaling graph sizes (CPU, median of 3 runs).

$ V $	$ E $	Analyse (s)	Graph→tensor (s)	HGT forward (ms)	Analyse : forward
249	1,127	0.27	0.011	22.5	12×
499	2,402	0.95	0.022	15.7	61×
999	6,422	4.74	0.055	36.7	129×
1,998	19,301	23.83	0.153	43.7	545×

The neural model is the cheapest stage, but not the whole cost. At 2,000 components the HGT forward pass takes 43.7 ms against 23.8 s for deterministic structural analysis. That ratio locates cost *inside* the pipeline; it is not the cost of evaluating an architecture. Indices 0–17 of every node feature vector (Section 3.4) — betweenness, closeness, reverse PageRank, articulation and bridge scores — are products of that same analysis stage, so the forward pass cannot run without it. End-to-end evaluation of an unseen 2,000-component architecture therefore costs about 24 s, of which the model is 0.2%; the 43.7 ms figure is the marginal cost of re-scoring an already-analysed graph. Cost scales with edge density rather than node count: the 520-component Enterprise mesh needs 27.2 s for its 3,245 structural edges, while a sparser 999-component system needs 4.7 s. Across the corpus the complete gate (structural analysis plus 18 anti-pattern detectors) runs in 0.02–27.4 s. Since the dominant stage is $O(|V|^2 + |V||E|)$, this budget should not be extrapolated: we have measured up to 2,000 components, and the systems invoked in Section 1.1 can be an order of magnitude larger.

The quantity measured is wall-clock time, against multi-seed discrete-event simulation sweeps that are substantially more expensive on the same corpus. We deliberately do not convert it into an energy or carbon figure (Section 8.2). What the measurements support is the narrow claim that pre-deployment gating fits inside a pull-request budget at the scales we tested, without a simulation cluster.

8. Discussion, Threats to Validity, and Conclusion

8.1. Discussion and Practical Implications

Role of the Predictive Pathway. Heterogeneous graph learning proves superior in three critical aspects: (1) *Zero-Shot Inductive Generalization:* It is the most effective *learned* model on unfamiliar architectures

($\rho = 0.664$ out-of-distribution); at matched QoS encoding the typed model leads its untyped counterpart by $+0.082$, and by $+0.064$ when neither encodes QoS, in both cases winning all twelve folds ($p = 0.0005$; Section 7.3). The margin is small — a third of Topo-QoS’s own $+0.253$ over unweighted centrality — but it is the only comparison in this study that wins every fold, under both edge channels and in both evaluation regimes. (2) *Shortlist Identification*: It attains the highest critical-set score of any variant we evaluated ($F_1@K = 0.439$), though its margin over the training-free QoS baseline is small and not statistically significant (7/12 folds, $p = 0.23$; Section 7.2), and it does not separate from untyped learning on this metric at all (-0.001 , $p = 0.32$). (3) *Low Marginal Cost*: Re-scoring an analysed graph costs 43.7 ms, and the complete gate including feature computation runs in 0.02–27.4 s — fast enough for pull-request gating rather than deferred nightly batch sweeps (Section 7.6). Beyond scalar rankings, HGT provides relation-specific attention over the channels driving cascade spread, per-relationship edge criticality, and multi-task quality attributions that degree centralities cannot produce.

Boundary Conditions and When Not to Train. The honest comparison is against a training-free QoS-weighted centrality baseline (Topo-QoS). On the synthetic folds the out-of-distribution ranking margin is $+0.058$ at fixed configuration ($p = 0.204$, Table 8) and $+0.115$ under nested selection ($p = 0.052$, Section 7.2.2) — neither reaching significance.

Off the corpus the result is not a tie but a loss. On the five open-source reference systems, scored against the oracle the model was trained on and against training-free references computed from the identical labels and node sets, HGT-QoS averages $\rho = 0.125$ where RM reaches 0.516 and Topo 0.511, going negative on two of the five (Section 7.5.1). This is the case the learned model was supposed to own — architectures structurally unlike anything in training — and it is the case where the baseline wins by the largest margin in the study. We take the two results together rather than separately: a learned ranker that does not separate from a closed-form structural score in distribution, and falls well behind it out of distribution, is not justified by ranking accuracy on this evidence.

The recommendation follows directly. A team that needs a component ranking should use Topo-QoS: no training, no corpus, no checkpoints, and better transfer to architectures it has never seen. Training is justified by what the baseline structurally cannot produce — relational attention over the channels carrying a cascade, per-relationship edge criticality, and multi-task quality attributions — and by one property the learned model does deliver on real systems: it signals its own unreliability. The $\hat{\sigma}$ diagnostic of Section 7.3.2, derived on synthetic folds, correctly rejects all five open-source architectures at a threshold calibrated on the corpus (Section 7.5.1). A predictor that can be trusted to say when it should not be trusted is deployable in a gate; one that is merely accurate on average is not. That, rather than a ranking margin, is where we would locate the learned pathway’s contribution on present evidence.

The continuous QoS encodings should be read in similarly measured terms. On the pre-repair corpus they did not improve ranking accuracy (-0.005 , $p = 0.68$) but halved seed-to-seed variance, which matters for a gate whose verdict should not depend on an initialisation. Whether the accuracy result survives the corpus repair of Section 7.1 is untested.

Role of the Explanation Layer. The RM profile separates single-point-of-failure exposure (Availability) from wide error propagation (Fault Tolerance), a distinction neither the predictor nor the oracle produces and one that decides between replication and circuit-breaking. Table 8 and Figure 3 therefore list RM alongside the ranking predictors as a diagnostic reference, not a competitor. Its practical value rests on that distinction being correct and useful, which this study specifies but does not evaluate against expert judgement or remediation outcomes.

8.2. Performance and Computational Sustainability Implications

The cost profile in Section 7.6 inverts the usual intuition about deep learning in continuous delivery: within SaG the neural model is the cheapest stage, and its margin widens with scale ($12\times$ at 249 nodes, $545\times$ at 1,998). Because feature extraction is a prerequisite rather than an alternative, the figure that matters operationally is the end-to-end one, and it is small enough for per-commit gating without a simulation cluster. Two qualifications bound this. First, the deterministic stage is $O(|V|^2 + |V||E|)$, so the corpus

measurements do not extrapolate to the very large deployments the introduction invokes: at the observed growth, a 10^4 -component system would leave the pull-request budget, and we have not measured one. Second, the quantity we measured is wall-clock time, not energy. No power counters were instrumented, the mapping from CPU-seconds to joules depends on hardware and utilisation we did not control, and a latency ratio is not an energy ratio.

On the sustainability argument. The broader claim available to a framework of this kind — that preventing cascading failures averts the retry storms, restart thrashing and connection-pool starvation that burn production capacity on uncompleted work — is a motivation for the research programme, not a result of this study. We did not deploy SaG, prevent an incident, or measure avoided compute. We state the mechanism because it is what would make the work matter for sustainability, and we mark it as unmeasured because it is.

8.3. Threats to Validity

Construct Validity. Our ground truth is generated via discrete-event cascade simulation on structural models rather than observing live production outages. To characterize construct divergence, we evaluated the three reliability-facing simulation engines (**FaultInjector**, **FailureSimulator**, **MessageFlowSimulator**) of our four-oracle taxonomy (Section 4.3). On *ordering*, the evidence converges strongly: the behavioural queue-flow simulator and the topological cascade oracle agree at $\rho = 0.907$ (never below 0.748 across scenarios), confirming that $I^*(v)$ tracks dynamic service disruption reach rather than purely static connectivity. However, queue simulation deltas are unweighted by message priority and only partially sensitive to delivery policies. On *critical-set membership*, agreement is lower: top- K Jaccard sits between 0.24 and 0.49 (against 0.111 by chance). Tie-handling accounts for part of this and we no longer claim otherwise: on the regenerated corpus the tie-robust cut shifts the mean Jaccard by up to 0.023 and a single scenario by as much as 0.167, where an earlier draft reported the shift as negligible. The residual reflects intrinsic non-linear cascade threshold sensitivity and oracle self-agreement noise (oracles achieve only 0.44 Jaccard against their own re-runs in worst-case scenarios). The related zero-inflation bound is reported in Section 7.4.6: $I^*(v)$ ties 19–106 Applications per scenario at exactly zero, and restricting agreement to components both oracles score positive moves individual scenarios by 0.2–0.3 in *both* directions — including **hub_and_spoke**, whose apparent sign disagreement ($\rho = -0.044$) becomes $\rho_{>0} = +0.263$ and is therefore a tie artifact rather than inverted ordering. A further 30–47% of components per scenario previously carried no simulated ground truth because the cascade model could not express direct Topic or Node failure; the injector now expresses both (a failed Topic loses its whole feed regardless of surviving publishers, and a failed host takes every component deployed on it via **RUNS_ON** simultaneously), and both strata are fully labelled and non-degenerate across all seven scenarios. The extension is additive — neither branch is reachable from an Application, Broker or Library injection — so every label those strata were scored on is unchanged and no table in Section 7 moves. Coverage is not informativeness, however, and we report each new stratum against the count baseline that would explain it away (subscriber fan-out for Topic, colocation count for Node): Spearman ρ against that baseline ranges -0.05 – 0.70 for Topic and 0.15 – 0.78 for Node, so the strata are not restatements of a degree count, but the comparison is reported rather than omitted. No learned model is trained on either stratum, so the typing result of Section 7.3 does not extend to them. The fourth oracle, $I_M(v)$ (**ChangePropagationSimulator**), targets Maintainability and traverses the same derived dependency topology from which $M(v)$ is scored; agreement between them serves as an internal consistency check rather than independent validation. Settling this requires an external referent independent of topology, such as version-control code churn or historical co-change coupling, left to future work.

Internal Validity and Substrate Confound. Potential feature leakage is prevented by strict graph view separation: predictors operate exclusively on G_{analysis} , whereas ground-truth simulators operate on $G_{\text{structural}}$. Two design changes made during development bear on how the shipped formulation should be read. First, an early hard articulation-point gate caused Availability $A(v)$ to collapse to a constant multiple of $w(v)$ in six of the eight scenarios then in the corpus where applications do not form cut vertices; replacing it with a continuous redundancy-deficit metric (fractional increase in average shortest paths upon removal) restored

structural discriminability, and Equation (16) is the post-change form. That diagnosis was made on the pre-recalibration code and, unlike every measurement in Section 7, is not backed by a regenerable artifact in the replication package; we record it as development history, not as a result. Second, we identified and resolved a normalization defect in the code-quality scoring pipeline: a min-max normalization helper previously assigned a maximal penalty to zero-variance populations, penalizing unanalyzed libraries lacking source-level `code_metrics`. Scoring undifferentiated populations with zero penalty corrected library Maintainability tiers while leaving Application rank correlations unaffected ($\Delta\rho \in \{-0.003, 0.000, 0.000\}$). In addition, the corpus on which Sections 7.2–7.4 were measured assigned an identical (RELIABLE, TRANSIENT_LOCAL) pair to every topic in seven of its twelve scenarios, giving Shannon entropy of exactly zero on all three QoS fields and leaving $w(t)$ standard deviation ≈ 0.02 there. Topo-QoS nevertheless gained $+0.319\rho$ over Topo in those scenarios (Table 6), a gain resulting from uniform edge rescaling and from Topic.criticality, which remained non-degenerate everywhere, rather than from QoS diversity. That defect has since been traced to the generator reading a name-pattern lookup instead of the declared per-scenario distributions, and repaired; the corpus that ships with this paper has *zero* degenerate scenarios of twelve (Section 7.1). Results predating the repair are marked in their captions, and their QoS-related readings are provisional until re-measured — including the ablation of Section 7.4.1, whose null result is what an inert treatment produces and is not evidence that declared QoS is uninformative. Independently of the repair, three of the seven QoS edge dimensions (`has_deadline`, `deadline_ns_log`, `max_blocking_ms_log`) remain identically zero in all twelve scenarios, because QoSPolicy carries no such fields for the generator to emit; populating them is a precondition for testing the per-dimension encoding claim at all, and we flag it as required future work rather than as a settled negative finding. As established in Section 6.2.1, both learned models operate on the shared native multigraph in both evaluations, so the out-of-distribution margin carries no multi-entity visibility confound. Three further confounds *were* present in an earlier version of the harness and are quantified in Section 7.3.1: the homogeneous branch trained on one scenario rather than $N - 1$, depth varied with which scenario was held out, and checkpoints were selected on a split of a training graph. Correcting them moves the untyped baseline from $\rho = 0.115$ to 0.600 and reduces the typed margin from $+0.246$ to $+0.082$. We report the corrected figure throughout and retain the attribution table so that the discrepancy against the shipped replication package is explicable rather than mysterious. The residual margin still combines two factors — GAT-N-QoS’s edge channel carries only the scalar aggregate $w(e)$ whereas HGT-QoS consumes all 16 dimensions — but the QoS ablation of Section 7.4.1 now attributes *none* of it to the QoS dimensions (-0.005 , $p = 0.68$), leaving typing itself as the operative factor. Contrary to what we previously reported, the typed advantage is not confined to distribution shift: it is present in-distribution as well ($+0.088$, $6/7$, $p = 0.047$), so relational typing should be understood as a consistent small benefit in both regimes rather than as a purely inductive bias. Furthermore, as disclosed in Section 8.2, sustainability claims rest on CPU latency reductions rather than direct hardware power counters (RAPL, NVML).

External Validity and Real-World Metric Stratification. While our evaluation spans twelve architectures across eight domains, including five authentic open-source distributed systems (ROS 2 Autoware, Cloud Microservices, Train-Ticket, Home Assistant, EdgeX Foundry), future work should evaluate larger enterprise deployments with hundreds of microservices. Crucially, as shown in Table 14, evaluating top- K critical sets across the pooled multigraph yields elevated $F_1@K$ scores (0.667–1.000) because unmeasured infrastructure entities carry constant zero failure labels. Reporting stratified metrics on active application services ($F_1@K = 0.000$ –0.625) provides the true, non-inflated operational baseline.

Conclusion Validity. Given the heavy-tailed distribution of cascading failure impacts, statistical comparisons utilize non-parametric rank correlation (Spearman ρ , Kendall τ), bootstrap confidence intervals ($B = 2,000$), and paired Wilcoxon signed-rank tests. Two critical hazards that altered conclusions in this study deserve explicit statement: (1) *Simpson’s Paradox*: Pooling rank correlations across node types moves them outside the envelope of the per-type correlations they aggregate, on both benchmarks we ran. Against $I_{\text{comp}}(v)$, RM’s per-type correlations of 0.503/0.395/0.142 pool to 0.028 (Section 7.4). Against $I^*(v)$ on the LOSO folds, pooling places RM at -0.014 and untyped GAT-N at 0.381, exactly reversing the standing that stratified evaluation gives them ($+0.195$ and 0.086). The two pooled figures are separate measurements against different

oracles and corpora, and are not comparable to each other. All reported metrics are therefore stratified on a single stated population (V_{app}). (2) *Substrate Conflation*: Early RM ablations scored $Q(v)$ from features restricted to the Application–Library `DEPENDS_ON` projection, where cut-vertex Availability terms vanished, producing false negative rank correlations. Computing features through the full multigraph pipeline resolved this issue.

8.4. Limitations and Future Work

Distributed AI and LLM Serving Topologies. Modern AI infrastructure relies on distributed LLM serving systems (e.g., vLLM, Triton, DeepSpeed) characterized by complex tensor and pipeline parallelism across GPU nodes, dynamic KV-cache routing, and disaggregated prefill-decode architectures. A straggler or failing node in a pipeline-parallel ring induces severe head-of-line blocking and massive GPU idle power dissipation. Extending the SaG multigraph schema to model distributed model serving topologies (nodes representing GPU workers, edges encoding tensor communication channels and KV-cache transfer fabrics) offers a promising avenue to forecast and mitigate bottlenecks in AI serving clusters.

Empirical Power and Hardware Testbeds. Validating the sustainability thesis through empirical power measurements using running package counters (Intel/AMD RAPL, NVIDIA NVML) and IPMI sensors on Kubernetes clusters during live fault injection, alongside Hardware-in-the-Loop (HIL) validation on cyber-physical testbeds (e.g., ROS 2 CAN-bus autonomous driving compute boxes).

Automated Refactoring and Self-Healing. Extending SaG from predictive analysis to prescriptive synthesis—automatically generating pull requests that reconfigure QoS policies, insert circuit-breakers, and add redundant broker pathways to eliminate single points of failure.

8.5. Conclusion

This work introduced **Software-as-a-Graph (SaG)**, a pre-deployment Static System Analysis framework that bridges the Architecture–Code Gap by combining a relation-specific Heterogeneous Graph Transformer (HGT) for failure-impact forecasting with an interpretable ISO/IEC 25010 Reliability–Maintainability explanation layer. Of the three concerns the framework is motivated by, this study delivers evidence on one and a half: **reliability**, through inductive cross-topology failure blast-radius forecasting on unseen architectures, and **analysis cost**, through a complete pre-deployment gate measured in seconds rather than cluster-hours. **Performance** and **energy** are motivating concerns we do not measure — no latency or throughput quantity is predicted, and no power counter was instrumented.

Our empirical results across synthetic systems and five open-source reference systems (ROS 2 Autoware, Cloud Microservices, Train-Ticket, Home Assistant, EdgeX Foundry) show that typed learning over the full multigraph generalizes out of distribution better than untyped learning at matched training set, depth and model selection: $\rho = 0.664$ vs. 0.600 without QoS edge features and 0.659 vs. 0.577 with them, 12 of 12 folds in both cases ($p = 0.0005$). In-distribution the same ordering holds at $+0.088$ (6/7, $p = 0.047$), so typing is a small consistent benefit in both regimes rather than a purely inductive bias. Because every fold in this comparison is drawn from a single parameterized generator, the typing claim is established on synthetic architectures alone. Tested off the corpus it does not hold up: on five open-source reference systems the trained model averages $\rho = 0.125$ against 0.516 and 0.511 for two training-free references scored on identical labels, going negative on two of the five (Section 7.5.1). What does transfer is the model’s own reliability signal — the label-free $\hat{\sigma}$ diagnostic correctly rejects all five. We are equally explicit about the boundary: against an unparameterized QoS-weighted centrality baseline (**Topo-QoS**), out-of-distribution ranking is not beaten — $\Delta\rho = +0.058$ ($p = 0.204$) at fixed configuration and $+0.115$ ($p = 0.052$) under nested selection — and the critical-set margin ($F_1@K = 0.439$ vs. 0.375) is not significant either (7 of 12 folds, $p = 0.23$). On present evidence a team needing only a ranking should use the baseline, in distribution and out of it. The case for the learned model rests on what the baseline cannot produce — relational attention, per-relationship criticality, standards-grounded attribution — and on its capacity to flag the architectures where it should not be trusted, which is the property that makes a pre-deployment gate safe to automate. Quantifying the first three is the natural next step; so is re-measuring Sections 7.2–7.4 on the repaired corpus (Section 7.1).

SaG establishes the typed substrate on which both prediction and explanation can be posed, providing an empirical and architectural foundation for dependable distributed systems engineering.

Declarations

CRedit Authorship Contribution Statement. **FIRST-NAME LAST-NAME:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **FIRST-NAME LAST-NAME:** Conceptualization, Methodology, Validation, Writing – review and editing, Supervision, Project administration.

Declaration of Competing Interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability. The complete replication package—including synthetic scenario datasets, generator configurations, simulation harnesses, real-world architecture adapters, trained model checkpoints, and all analysis scripts—is deposited in a public archive with a persistent identifier and cited as [85].

Declaration of Generative AI in the Writing Process. During the preparation of this work, the authors used AI-assisted language tools to check grammar, improve readability, and support LaTeX typesetting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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